Emulator of PR-DNS: Part II, Accelerating thermodynamics and cloud droplet fields with machine learning in Particle-Resolved Direct Numerical Simulation

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Abstract

Particle-resolved direct numerical simulations (PR-DNS) are crucial for unraveling the intricate interplay of aerosol-cloud-turbulence processes. However, such models are challenged by the huge computational cost due to the extremely high resolution. Our prior work showcased that leveraging machine learning emulators could slash computational expenses by two orders of magnitude while maintaining remarkable precision for dynamic fields, and exhibited generalizability across diverse initial conditions and at super-resolution scales without retraining the emulators. Building upon this foundation, this work extends the emulator's application to thermodynamic in the two spatial dimensions and droplet fields in the three spatial dimensions. Furthermore, to enhance the robust generalizability of the emulator for different initial values and super resolution, we introduce a novel multi-initial learning approach for the neural operator method. For the droplet fields, we introduce a novel loss function tailored to assess distribution differences using the Mallows distance, focusing particularly on droplet size distributions. Our findings indicate that the machine learning emulators hold promising potential to effectively mimic numerical PR-DNS simulations, thereby significantly advancing our understanding of the complex interactions within aerosol-cloud-turbulence processes.

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Key Points:

- The Fourier Neural Operator can effectively emulate the moist process, expanding up our prior work for emulating dynamics and temperature.
- The multi-initial learning method enhances the generalizability across diverse initial conditions and super resolutions.
- The droplet radius distribution can be learned using the Mallows distance.

Plain Language Summary

Understanding the aerosol-cloud-turbulence interaction is crucial for climate and weather prediction. However, simulating these processes using extra-high-resolution particle-resolved direct numerical simulations is computationally expensive. In our previous work, we demonstrated that using machine learning emulators can significantly reduce computational costs while maintaining high accuracy. In this study, we extend the emulator to thermodynamic fields in two dimensions and droplet behavior in three dimensions. For the emulator of thermodynamic fields, we improve the generalizability across various initial conditions and resolutions without requiring retraining the machine learning models. For the emulator of droplet fields, we introduce a novel loss function to evaluate the droplet size distribution. Our findings indicate that these machine learning models could serve as effective tools for more efficient simulation of intricate aerosol-cloud-turbulence interactions.

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Abstract

Particle-resolved direct numerical simulations (PR-DNS) are crucial for unraveling the intricate interplay of aerosol-cloud-turbulence processes. However, such models are challenged by the huge computational cost due to the extremely high resolution. Our prior work showcased that leveraging machine learning emulators could slash computational expenses by two orders of magnitude while maintaining remarkable precision for dynamic fields, and exhibited generalizability across diverse initial conditions and at super-resolution scales without retraining the emulators. Building upon this foundation, this work extends the emulator's application to thermodynamic in the two spatial dimensions and droplet fields in the three spatial dimensions. Furthermore, to enhance the robust generalizability of the emulator for different initial values and super resolution, we introduce a novel multi-initial learning approach for the neural operator method. For the droplet fields, we introduce a novel loss function tailored to assess distribution differences using the Mallows distance, focusing particularly on droplet size distributions. Our findings indicate that the machine learning emulators hold promising potential to effectively mimic numerical PR-DNS simulations, thereby significantly advancing our understanding of the complex interactions within aerosol-cloud-turbulence processes.

1 Introduction

Understanding the intricate interplay within the aerosol-cloud-turbulence system is critical for accurately simulating the impacts of climate change and extreme weather events (Liu, 2019; Fan et al., 2016; Simpkins, 2018). Despite the advances of General Circulation Models (GCMs), Cloud-Resolving Models (CRMs), and Large Eddy Simulations (LES), uncertainties persist even at the high resolutions offered by LES. This is primarily attributed to the existing knowledge gaps and uncertainties in sub-grid parameterizations. Even for sub-LES scales. critical processes such as turbulence-microphysics interactions, turbulent entrainment-mixing, and droplet clustering are either inadequately represented or not at all in these models (Liu et al., 2023). This limitation significantly hampers the advancement in simulating the aerosol-cloud-turbulence system (Hourdin et al., 2017; Khairoutdinov & Kogan, 2000; Yang et al., 2012).

Particle-resolved Direct Numerical Simulation (PR-DNS) eliminates the need for sub-grid parameterizations (Chen et al., 2018; Liu, 2019). It explicitly resolves the smallest turbulent eddies for fluid dynamics and thermodynamics, as well as individually tracks the motion and growth of each droplet (Kumar et al., 2012, 2013). Within our current PR-DNS model, we employ the point-particle approximation, including essential droplet properties such as radius, position, and velocity. The droplet behavior is intricately intertwined with the dynamics and thermodynamics of the surrounding turbulent fluid (Gao et al., 2018). The PR-DNS serves as a unique and valuable tool for advancing our fundamental understanding of the process-level process of the aerosol-cloud-turbulence system. However, the extreme high resolution and tracking of individual droplets impose a significant computational cost. This limitation presently restricts its utility to simulations within smaller domains and a limited number of physical processes (Gao et al., 2018).

Recently, a line of studies use deep neural networks to build emulators to simulate turbulent fluid flows, which can significantly reduce the computational cost compared with traditional numerical solvers (Bhatnagar et al., 2019; Kochkov et al., 2021). However, neural network-based solvers often lack generalizability due to their finite-dimensional mapping. The neural operators, especially the Fourier Neural Operator, do the infinite-dimensional mapping, which map inputs to the Fourier space, learn the transformations, and subsequently reverse them back to the spatial domain (Li et al., 2020a, 2020b). Our previous research demonstrates that the neural operator emulators can achieve high accuracy, and significantly reduce computational cost by two orders of magnitude, and exhibit generalization across diverse initial conditions and super-resolution simulation without retraining the machine learning models. However, our findings have primarily focused on fluid dynamics and the temperature field (Zhang et al., 2024; Atif, López-Marrero, et al., 2024; Atif, Dubey, et al., 2024).

In this study, we expand upon our prior research to encompass the moist process, with foci on both water vapor/ supersaturation fields and microphysics/droplet size distributions. In terms of generalizability across different initial conditions and super-resolution simulations, our previous work did not provide robust generalizability because the fundamental model was trained on only a single set of initial conditions from the PR-DNS simulation. This narrow training approach could not ensure optimal performance on unseen data arising from different initial conditions. To achieve truly robust generalizability, models need to be trained on diverse PR-DNS simulations spanning a wider range of initial conditions. We thus introduce a novel multi-initial learning approach wherein our neural operator emulators are trained using multiple PR-DNS simulations with diverse initial conditions.

Furthermore, to learn the droplet size distribution, FNO cannot be directly applied. The primary challenge lies in devising a suitable loss function to measure the distance between the predicted and true distributions. While Kullback-Leibler (KL) divergence is commonly used to measure differences between probability distributions, it requires the two distributions to have identical bin structure (Kullback, 1997). However, the corresponding bin placements in the true distributions from PR-DNS and those from emulators may not align perfectly. Therefore, we design a novel loss function customized to assess distribution disparities using the Mallows distance (Levina & Bickel, 2001). Our results indicate that the new neural operator demonstrates high accuracy in both Euler dynamics, thermodynamics, and Lagrangian droplet radius distributions. This tool is pivotal for advancing PR-DNS in larger domain simulations with reduced computational costs, thereby playing a crucial role in deepening our understanding of the intricate interactions within aerosol-cloud-turbulence systems.

The remainder of the paper is structured as follows. Section 2 presents the PR-DNS model and the simulation data utilized for training the emulators. Section 3 outlines the methodologies employed, including the neural operator, multi-initial learning, and the use of the Mallows distance as a loss function for learning the droplet radius distribution. Section 4 delves into the performance evaluation of the methods. Conclusions and discussions are provided in Section 5.

2 Model and Data

The PR-DNS model incorporates the Eulerian field representation of turbulence near the Kolmogorov microscale and Lagrangian droplet tracking. Comprehensive details regarding the PR-DNS can be found in (Gao et al., 2018) and (Zhang et al., 2024). Here, we provide a brief overview of the key equations.

The Eulerian dynamic field is the incompressible Boussinesq fluid system, described by

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_0}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}_b + \mathbf{f}_e \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{1b}$$

where $\mathbf{u}$ is the velocity field, p is the pressure field, ν is the kinematic viscosity, ρ_0 is the density of dry air, $\mathbf{f}_b$ and $\mathbf{f}_e$ are the buoyancy and external forcing, respectively.

The Eulerian thermodynamic fields, namely the temperature and vapor mixing ratio, are given by

$$\partial_t T + (\mathbf{u} \cdot \nabla)T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T \tag{2}$$

$$\partial_t q_v + (\mathbf{u} \cdot \nabla)q_v = -C_d + \mu_v \nabla^2 q_v \tag{3}$$

where L_h is the latent heat of water vapor condensation, c_p is the specific heat, μ_T and μ_v are the molecular diffusivity for temperature and water vapor, respectively. The condensation rate C_d depends on the droplet population, thereby establishing a coupling between the Eulerian field and the Lagrangian droplets.

For the Lagrangian process, the motion of droplets is calculated by,

$$\frac{\mathrm{d}\mathbf{X}_i(t)}{\mathrm{d}t} = \mathbf{V}_i(t) \tag{4a}$$

$$\frac{\mathrm{d}\mathbf{V}_i(t)}{\mathrm{d}t} = \frac{1}{\tau_p}\left[\mathbf{u}\left(\mathbf{X}_i, t\right) - \mathbf{V}_i(t)\right] + \mathbf{g} \tag{4b}$$

where $X_i(t)$ and $V_i(t)$ are the position and velocity of the ith droplet at time t. The droplet velocity is influenced by turbulence. τ_p is the droplet response time, which measures the inertial effect. The gravity term $\mathbf{g}$ represents the sedimentation, which affects the vertical motion of droplets.

The growth of droplet condensation is described

$$R_i(t)\frac{\mathrm{d}R_i(t)}{\mathrm{d}t} = A \cdot S\left(\mathbf{X}_i, t\right) \tag{5}$$

where $R_i(t)$ represents the radius of the ith droplet at time t, A is a function of temperature and pressure. Further details can be found in (Gao et al., 2018). The supersaturation $S(X,t)$ is calculated as

$$S(\mathbf{X}, t) = \frac{q_v(\mathbf{X}, t)}{q_{v,S}(\mathbf{X}, t)} - 1 \tag{6}$$

where $q_{v,S}$ is the corresponding saturation water vapor mixing ratio.

We leverage the PR-DNS simulations to generate high-fidelity dynamic, thermodynamic and droplet dataset. In the PR-DNS simulations to learn thermodynamics fields, the physical domain is set to be $0.512 \times 0.512 m^2$ in two spatial dimensions with doublly periodic boundary conditions. The training datasets are generated with 128 x 128 grid resolution, corresponding to a grid spacing of 4 mm. Additionally, to improve robust generalizability, we conduct simulations with 3 different initial velocity conditions at 128×128 grid resolution for training, reserving an additional case for testing. To evaluate the performance of predicting higher resolutions using the lower 128×128 training model, the test simulation resolution is 256×256, corresponding to a finer grid size of 2mm. The average time step size is 0.003s, satisfying the Courant–Friedrichs–Lewy stability criterion. The initial turbulent Reynolds and Rayleigh numbers are 6,827 and 40,227, respectively. Each simulation is run for 6,000 time steps. For a single initial condition, we use the first 4000 time steps of the simulation for training. For multiple initial conditions, we combine the 12,000 time steps across three simulations with distinct initial conditions for training. In each case, the last 2000 time steps are reserved for model validation. In the PR-DNS simulations to learn the droplet size distribution, we use the three-dimensional spatial domain. Further details can be found in (Gao et al., 2018). The output time interval is 0.01s. A total of 863 output steps are generated, with the first 600 steps is used to train the emulator of the droplet size distribution, and the rest for testing.

3 Method

3.1 Fourier Neural Operator

For dynamic and thermodynamic variables, like velocity, temperature and water vapor, the emulator could be developed using conventional deep learning methods, such as Convolutional Neural Networks (CNN, LeCun et al. (1998)), which utilize the current timestep as input to predict their values in the subsequent timestep. However, this approach would have limitations. Using the CNN-like methods, the relationship between the input and output is predetermined by the network architecture, as illustrated in Figure 1 (a). This method would not be adaptable for scenarios where the output resolution varies, like in the case of super resolution.

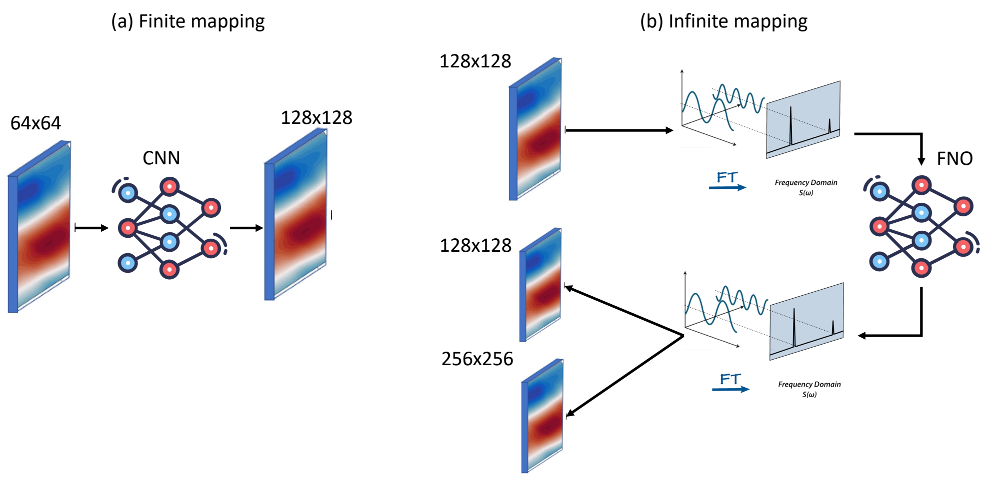

Figure 1. Differences in deep learning between finite mapping (a) and infinite mapping (b).

Instead, the Fourier Neural Operator (FNO) offers enhanced generalizability and the flexibility to adjust the output resolution. It transforms the spatial field into the Fourier domain. Subsequently, the neural network directly learns the mappings in Fourier space rather than in spatial space. Finally, it translates these representations back into the spatial domain, as shown in Figure 1 (b). Our previous work demonstrated that the FNO exhibited higher accuracy and efficiency compared to the finite mapping method for emulating both velocity and temperature fields (Zhang et al., 2024).

Details of the FNO algorithm can be found in Li et al. (2020a). Here, we provide a brief introduction to this algorithm. The FNO learning mapping equation is given by

$$\mathcal{G}_\theta = Q \circ \sigma_l \left(\mathcal{K}_l + W_l\right) \circ \cdots \circ \sigma_0 \left(\mathcal{K}_0 + W_0\right) \circ P \tag{7}$$

The neural operator can be decomposed into three components:

1. Encoder (P): It maps input spatial fields into a high-dimensional latent representation.
2. Fourier layers ($\mathcal{F}$): It is the foundational component of FNO, consisting of the integral operator (K_l) and the linear operator (W_l), where $l = 0, 1, 2, 3$, denoting the four Fourier layers.
3. Decoder (Q): It projects the Fourier output back to the original spatial field.

Both P and Q are shallow fully connected neural networks. In the integral operator (K_l), the input lifted by the encoder P first undergoes a fast Fourier transform (FFT) into Fourier space. Within this space, higher frequency noise is removed by truncating Fourier modes above a certain threshold. In our study, we find truncating above 30 modes achieves a balanced tradeoff between performance and computational efficiency. A linear neural network is then used to learn the remaining low-pass filtered Fourier coefficients. The result is reversed to spatial domain by an inverse FFT.

The output of each Fourier layer is derived from the combination of the integral operator (K_l) and linear operator (W_l), which is the 2-dimension convolution layer with Gaussian Error Linear Unit (GELU) activation function. The output is then fed into the subsequent layer. Finally, the cumulative is projected back onto the desired data dimension by the decoder network (Q).

3.2 Learning from multi-initial simulations

In our prior study (Zhang et al., 2024), we trained the FNO using PR-DNS simulations from a single set of velocity initial conditions and then applied it to obtain predictions given different initial conditions and also for higher resolution predictions to assess generalizability. However, training FNO with a single initial condition was found to lack robustness when applying the trained model to other initial conditions, as discussed in Section 4. To ensure a more robust model, we opt to train the FNO using data from multiple PR-DNS simulations conducted under a range of initial conditions.

Specifically, we generate simulations using four distinct velocity initial conditions with a 128 x 128 grid, as illustrated in Figure 2. They are generated by the Rogallo method (Rogallo, 1981), which a random factor is incorporated to generate the different initial conditions. The FNO model is subsequently trained on simulations from the first three initial conditions, and the fourth is used to test the performance, as depicted in Figure 3. By leveraging data from diverse initial conditions during training, the FNO method aims to learn the general turbulent flow dynamics rather than specific initialization-dependent behaviors. This improves the generalizability compared to our previous single-initialization FNO methodology, as will be shown in Section 4.1.

3.3 Predicting droplet size distribution with the Mallows loss

Cloud droplet size distribution is an important cloud microphysical property that impacts cloud processes and interactions within the climate system. PR-DNS can explicitly resolve the time-evolving droplet size distribution. However, building such a emulator presents challenges compared to the fluid dynamics fields. Unlike spatial grid-by-grid fields, the droplet size distribution is defined as a time series histogram, representing the frequencies within discrete radius bins at each time step:

$$\begin{aligned} h_{f_t} &= \left\{ \left([\underline{I}, \bar{I}]_{f1}, p_{f1} \right), \left([\underline{I}, \bar{I}]_{f2}, p_{f2} \right), ..., \left([\underline{I}, \bar{I}]_{fl}, p_{fl} \right) \right\}_t \\ h_{g_t} &= \left\{ \left([\underline{I}, \bar{I}]_{g1}, p_{g1} \right), \left([\underline{I}, \bar{I}]_{g2}, p_{g2} \right), ..., \left([\underline{I}, \bar{I}]_{gl}, p_{gl} \right) \right\}_t , \end{aligned} \tag{8}$$

where h_{f_t} and h_{g_t} are defined as the two histograms at the t-th time step, and h_{f_t} represents predictions and h_{g_t} represents the ground truth, $[\underline{I}, \bar{I}]_i$ represents the lower and upper bounds of each histogram bin, p_i is the frequency corresponding to the bin $[\underline{I}, \bar{I}]_i$, and fl and gl are the number of bins in the two histograms, respectively. The cumulative probability of the two distributions f and g are w_{f0},w_{f1}, ..., w_{fl} and w_{g0},w_{g1}, ..., w_{gl}, which are calculated by

$$w_i = \sum_j^i p_j, \tag{9}$$

where $i \in [f0, f1, ..., fl]$ or $[g0, g1, ..., gl]$. We use a simple regression model to predict each bin range and its corresponding frequency, described using PyTorch code as follows:

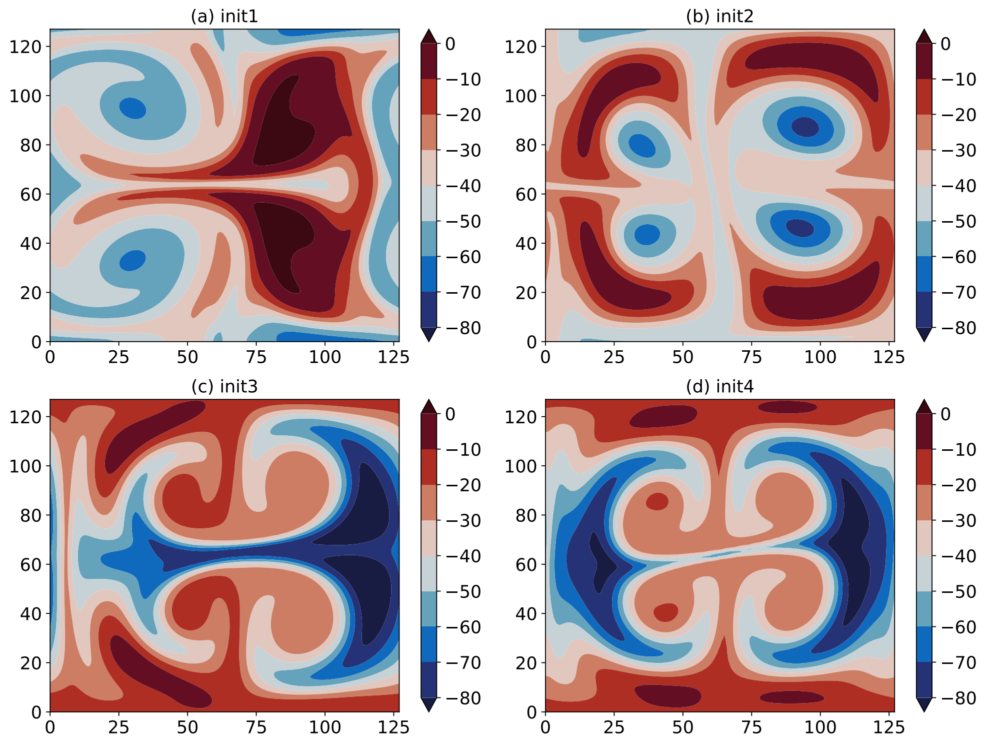

Figure 2. The time snapshot at the 100th step of supersaturation for the four different initial conditions.

```
class NN(nn.Module):
    def __init__(self,input_size, output_size):
        super(NN,self).__init__()
        self.con1 = nn.Parameter(torch.tensor(1.0))
        self.con2 = nn.Parameter(torch.tensor(-0.2))
        self.con3 = nn.Parameter(torch.ones(200))

    def forward(self, bin_input,freq_input):
        bins = bin_input[:,0,:] * self.con1 + self.con2
        freq = freq_input[:,0,:] * self.con3
        bins = torch.where(bins<0,0,bins)
        freq = torch.where(freq<=0,0,freq)
        return bins,freq
```

To build the emulator of droplet radius histogram time series, it is essential to define an appropriate loss function to measure the difference between the predicted and true distributions. While KL divergence can serve as the loss function, it may not be suitable because of the mismatched bin structures between the two distributions. Since the parameters in the algorithm above require tuning through backward optimization. During the initial iterations, the bin structures may not align well. Instead, the Mallows method can evaluate distribution dissimilarity even when the bin structures differ (Mallows, 1972). The Mallows distance $D_M(f,g)$ is defined as follows and is represented by the shaded area in Figure 4:

$$D_M(f,g) = \sqrt{\sum_{z=1}^{m} \left(H_f^{-1}(z) - H_g^{-1}(z)\right)^2}, \tag{10}$$

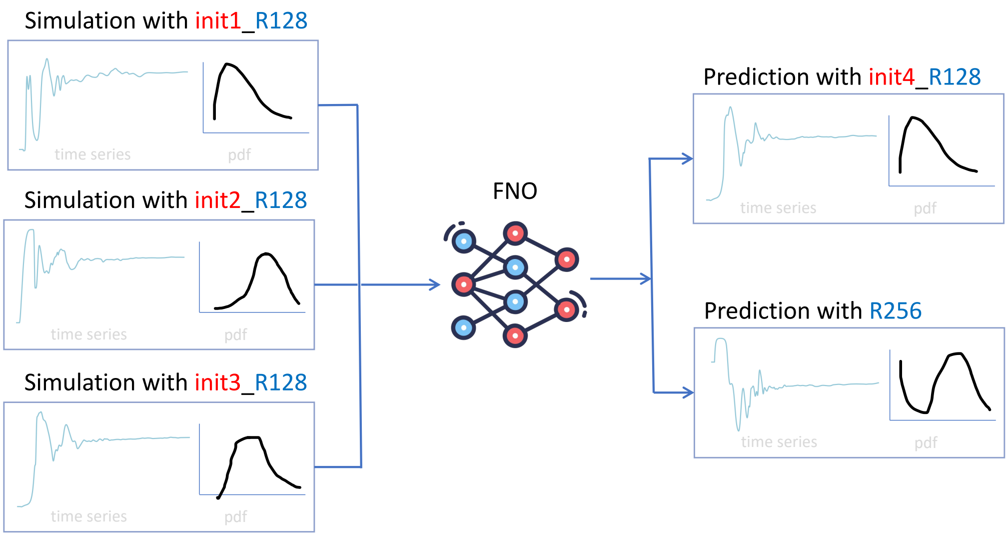

Figure 3. Illustration diagram of training FNO model with multi-initial simulations for different initial conditions and super resolution.

where $H_f^{-1}(t)$ and $H_g^{-1}(t)$ with are the inverse cumulative distribution functions of f and g, respectively, which can be defined by

$$\begin{aligned} H^{-1}(z) &= \underline{I}_l + \frac{z - w_{l-1}}{w_l - w_{l-1}} \left(\bar{I}_l - \underline{I}_l\right) \\ &\text{if } z \in [w_{l-1}, w_l), \end{aligned} \tag{11}$$

It is worth noting that $H^{-1}(z)$ is a function with z as the single parameter. Both H_f^{-1} and H_g^{-1} utilize the same z value to calculate the difference. In Figure 4, z represents the values from the y-axis, obtained by combining the cumulative probabilities w for f and g, and then sorted without repetitions; m denotes the length of the z array.

Within the interval (z_{l-1}, z_l), it follows the uniform distribution. Hence, the inverse cumulative distribution can be expressed as:

$$H^{-1}(t) = c_l + r_l(2t - 1) \quad z_{l-1} \leq t \leq z_l \tag{12}$$

where the c_l and r_l are the center and radius of each bin l. The center of a bin refers to the midpoint of the bin. The radius of a bin is half of the width of the bin. To obtain the center and radius, the lower bound ($\underline{I}$) and upper bound ($\bar{I}$) for the interval (z_{l-1}, z_l) can be derived from the inverse cumulative function $H^{-1}(z_{l-1})$ and $H^{-1}(z_l)$, respectively. They can subsequently be calculated as follows

$$c_l = \frac{\underline{I}_l + \bar{I}_l}{2} \quad \text{and} \quad r_l = \frac{\bar{I}_l - \underline{I}_l}{2} \tag{13}$$

The Mallows distance in Eq. 10 can be rewritten as follows according to (Arroyo & Maté, 2009) and (Dias & Brito, 2015)

$$\begin{aligned} D_M(f,g) &= \sum_{l=1}^{m} \sqrt{\int_{z_{l-1}}^{z_l} \left(H_f^{-1}(t) - H_g^{-1}(t)\right)^2 \, \mathrm{d}t} \\ &= \sum_{l=1}^{m} p_l \sqrt{\int_0^1 [(c_{lf} + r_{lf}(2t-1)) - (c_{lg} + r_{lg}(2t-1))]^2 \, \mathrm{d}t} \\ &= \sum_{l=1}^{m} p_l \sqrt{[(c_{lf} - c_{lg})^2 + \frac{1}{3}(r_{lf} - r_{lg})^2]} \end{aligned} \tag{14}$$

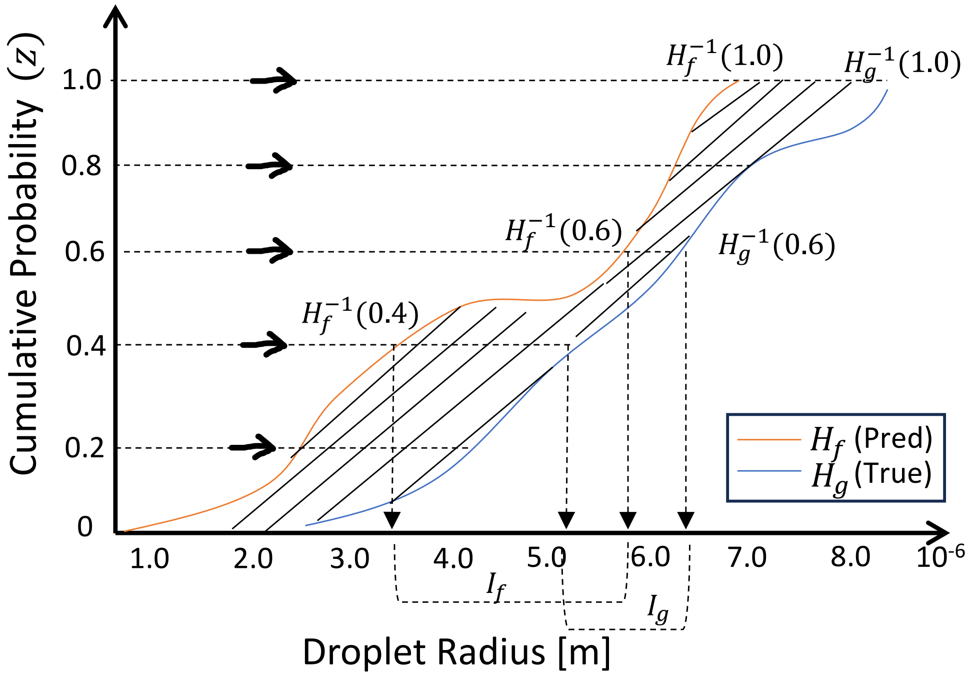

Figure 4. The Mallows distance quantifies the difference in droplet of the predicted and true cumulative radius distribution. The shaded area $\sqrt{\sum_{z=1}^{m}\left(H_f^{-1}(z) - H_g^{-1}(z)\right)^2}$ represents this distance.

4 Results

In this section, we assess the emulator performance of Eulerian thermodynamic fields and Lagrangian droplet radius distribution in predicting cloud microphysical properties.

4.1 Thermodynamics

4.1.1 Accuracy

In our prior research (Zhang et al., 2024), we conducted a comparative evaluation of three machine learning approaches for predicting dynamic and temperature fields: the UNet, ResNet, and FNO. The FNO exhibited the lowest prediction errors while achieving powerful generalizability. Therefore, in this current work, we focus on results using the FNO model alone.

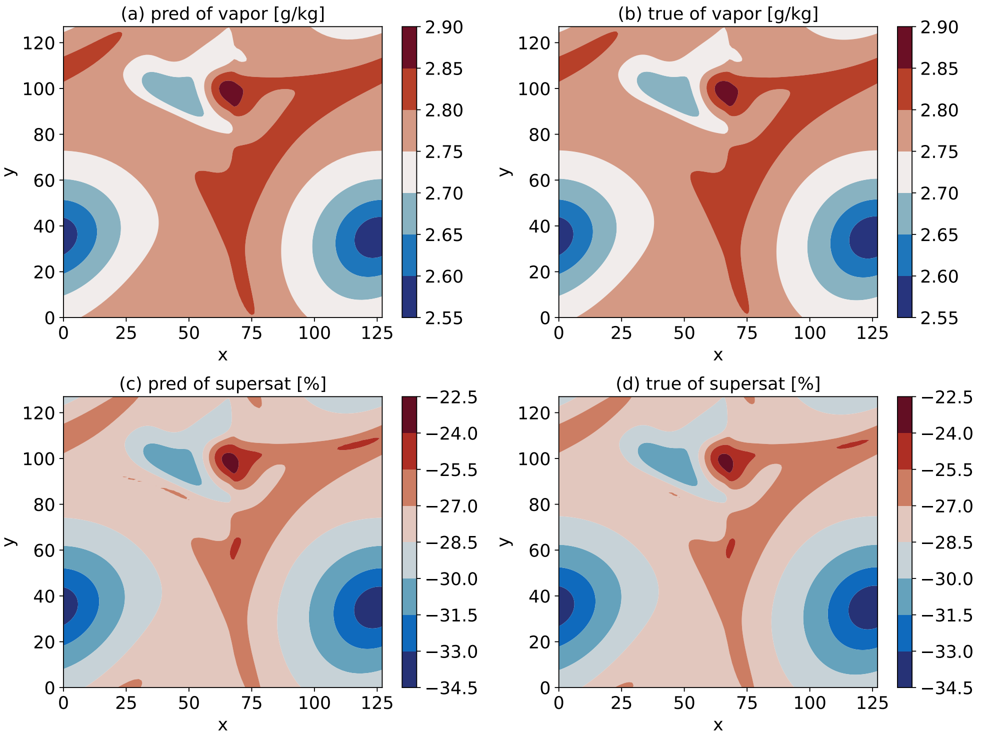

Figure 5. Snapshot predictions (a,c) and ground truth (b,d) of mixing ratio and supersaturation at the 5900th time step (equivalent to the 1900th time step in the testing dataset. The following figures about snapshot are all from the 5900th time step) with a resolution of 128 × 128.

Figure 5 displays the two-dimensional (2D) spatial maps of mixing ratio and supersaturation at a 128 x 128 resolution from a single time snapshot, as examples. The figure shows that the FNO predictions very closely resemble the ground truth obtained from the PR-DNS simulation for the two thermodynamic variables.

To further quantitatively evaluate the accuracy of the deep learning approach, Figure 6 show the coefficient of determination R^2 between FNO predictions and ground truth, evaluated over the entire testing dataset on a grid point by grid point basis. It represents the fit goodness through the proportion of variance. The value 1.0 of R^2 represents a perfect fit, and the value can range from negative to 1.0. The higher the value, the better the performance. The R^2 accuracy metric is calculated as follows,

$$R^2(x_i, y_j) = 1 - \frac{\sum_{k=1}^{T} \left(u(x_i, y_j, t_k) - \hat{u}(x_i, y_j, t_k)\right)^2}{\sum_{k=1}^{T} \left(u(x_i, y_j, t_k) - \bar{u}(x_i, y_j)\right)^2} \tag{15}$$

where x_i and y_j represent grid points for the spatial coordinates in the 2D space. T is the total length of the testing dataset (corresponding to the total number of testing time steps $\{t_k\}_{k=1}^{T}$ of time coordinates). $\hat{u}$ is the prediction by deep learning models, u denotes the corresponding truth, and $\bar{u}$ is the temporal average of u.

The spatial maps and error analyses demonstrate that the FNO emulators can accurately reproduce the original PR-DNS simulation in terms of the mixing ratio and supersaturation as well. Because the patterns of mixing ratio and supersaturation are similar, we will focus on the supersaturation field in the subsequent analysis.

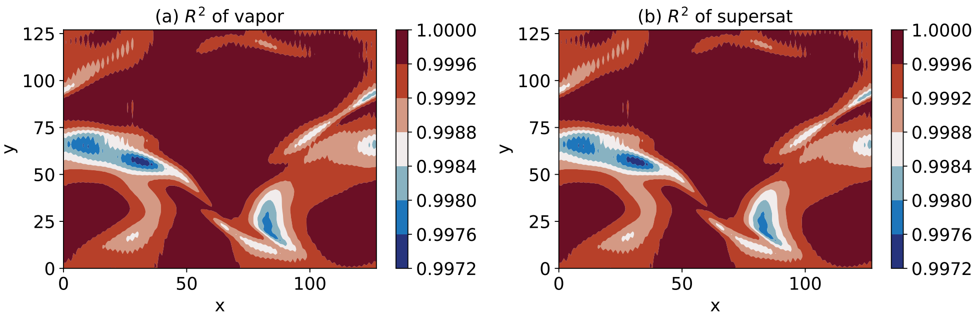

Figure 6. R^2 accuracy metric of mixing ratio (a) and supersaturation (b) between FNO predictions and the ground truth for the entire testing data set on the grid point by grid point with the resolution of 128 × 128, according to Equation 15.

4.1.2 Generalizability for different initial conditions and resolutions

While we have previously demonstrated the generalizability of our approach to different velocity initial conditions and higher resolutions, directly applying a model trained using one set of initial conditions to another set may introduce biases. We perform four PR-DNS simulations with different initial conditions, shown in Figure 2. Figure 7 (d) illustrates the case where the model trained on Init1 is applied to the Init4 conditions. The predictions exhibit a positive bias compared to the true values, suggesting the model has overfitted. Due to the magnitude similarity between init1 and init2, this bias is reduced when the model trained on init1 is applied to the init2 case, as shown in Figure 7 (b).

To address the issue, we combined data from the PR-DNS simulations using Init1, Init2 and Init3 to form a single, augmented training dataset. The FNO model is then trained using this combined dataset to learn patterns across the different data characteristics, rather than specializing to the single data. We evaluate the generalization of this novel training method on the independent Init4 condition. Figure 8 shows the greatly improved agreement between the prediction and the ground truth for the supersaturation field at the one time snapshot. This suggests that the model can effectively capture the features despite variations in the initial profiles. Figure 8 (c) confirms this method yields skillful predictions across the entire Init4 simulation based on the R^2 accuracy using the metrics defined by equation 15.

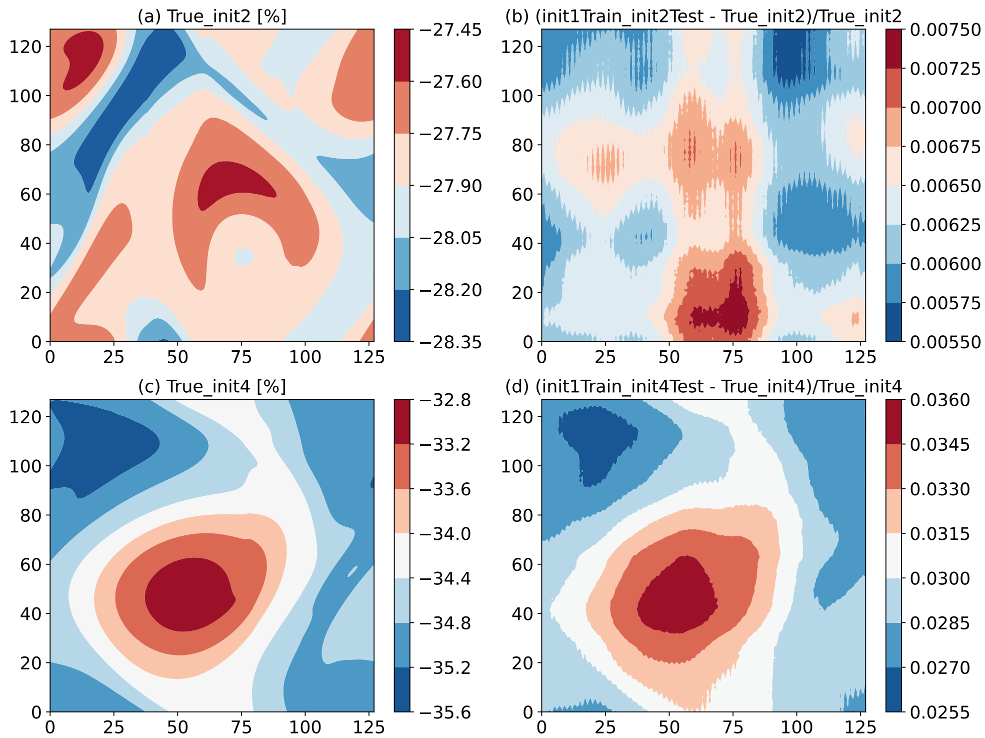

Figure 7. Testing the model trained on Init1 with Init2 and Init4, in terms of the ground truths and their relative predictive errors.

To assess the predictive capacity of super resolution using the emulator trained by the lower resolution dataset, we train the FNO model using PR-DNS simulations at a 128 x 128 resolution and apply it to predict fields at a higher resolution of 256 x 256. However, directly using the FNO model trained on one set of 128 x 128 resolution data could lead to biased predictions. In Figure 9, it is confirmed that training the FNO model using init1 at a 128 x 128 resolution and applying it to the 256 x 256 resolution results in a positive bias, as shown in Figure 9 (c). This discrepancy arises because the training data from init1 lower resolution simulation systematically exceeds the testing data from the higher resolution simulation, as illustrated in Figure 9 (a-b).

To address this challenge, we once again utilize the Multi-Init training approach. Figures 10 (a-b) shows the outcomes when employing this Multi-Init trained model to predict the 256 x 256 resolution in a single time snapshot. The predicted supersaturation field aligns closely with both the spatial patterns and the magnitude range of the actual field. Figure 10 (c) validates that this method produces accurate predictions throughout the entire Init4 simulation, as demonstrated by the R^2 accuracy metric.

4.2 Droplet Size Distribution

To assess the surrogate model's capability in predicting droplet size distribution, we employ the Mallows distance to define the loss function for the neural network surrogate model, a metric particularly suitable for comparing various histograms. Due to the subtle and gradual evolution of radius distributions over short time intervals, we extend the evaluation of predictions to longer timeframes ranging from 5 to 30 timesteps.

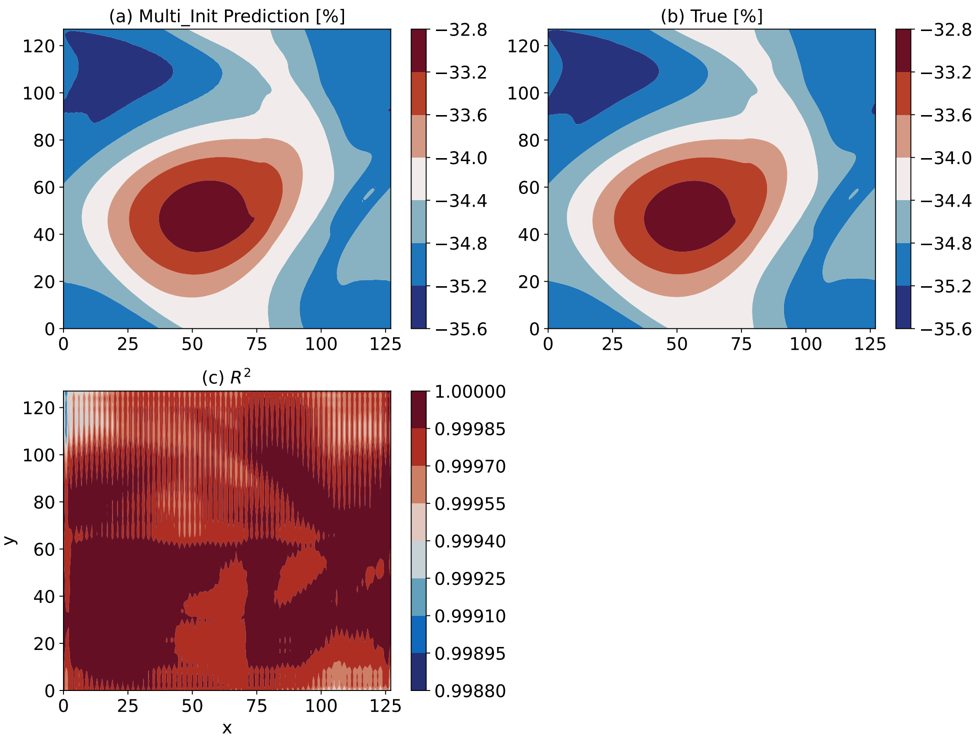

Figure 8. Multi-Init learning from the PR-DNS simulations with Init1, Init2, and Init3, testing on Init4 in terms of the time snapshot of predictions (a) and the ground truth (b), along with the R^2 accuracy (c).

Figure 11 illustrates that the Mallows loss rises as the forecast lead steps increase stabilizing after 20 steps (Here, the lead step refers to how far ahead the ML model is trying to predict. For example, if the ML model takes the current time step as input and predicts the values at the 7th subsequent step, the lead step would be 7). This indicates that the surrogate model retains its skill in simulating distribution changes, even as small uncertainties accumulate over longer time intervals. This observation is reinforced by qualitatively comparing the predicted and actual radius probability density fields at selected short and long lead times. The overall patterns remain comparable between true and predicted distributions, even out to the 30-step forecast horizon, as depicted in Figure 12.

We compare the regression method based on the Mallows loss function, as described in Section 3, with the conventional machine learning method, XGBoost. Two XGBoost models are configured: one for predicting the bins and the other for predicting the frequencies at the future step. Both XGBoost models take the bins and frequencies at the current step as input. In contrast to XGBoost, the Mallows loss function concurrently considers both aspects. Illustrated in Figure 13, the separate predictions in XGBoost result in a non-smooth distribution and significantly higher bias compared to the ground truth. Conversely, the Mallows loss function generates a smoother distribution with reduced bias.

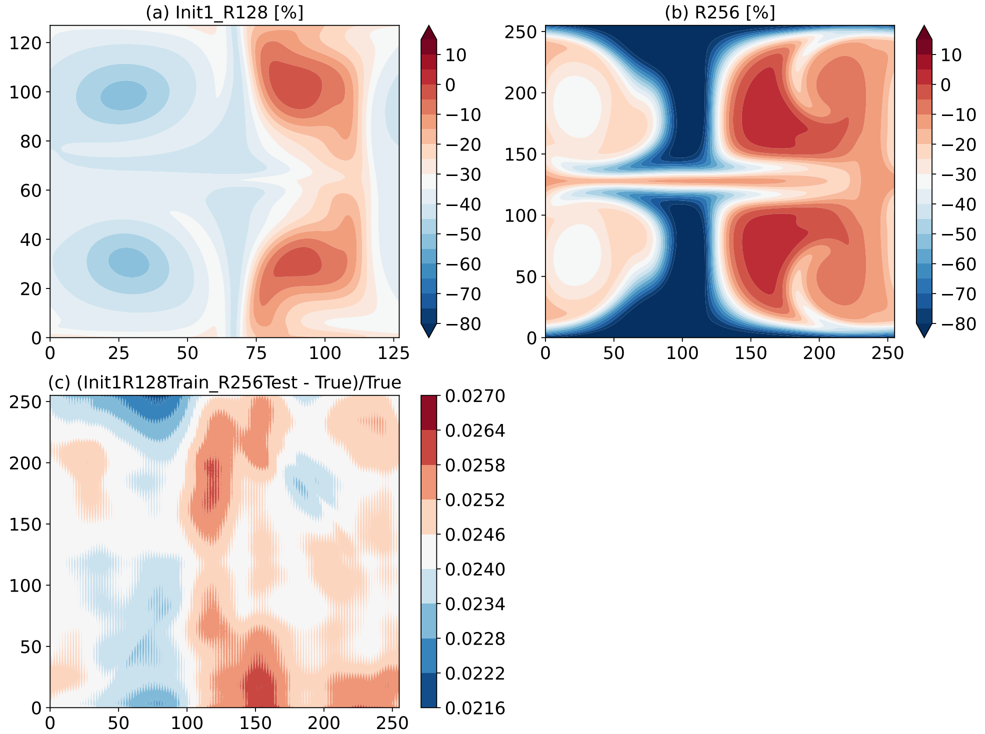

Figure 9. Testing the model trained on Init1 128 x 128 resolution with 256 x 256 resolution.

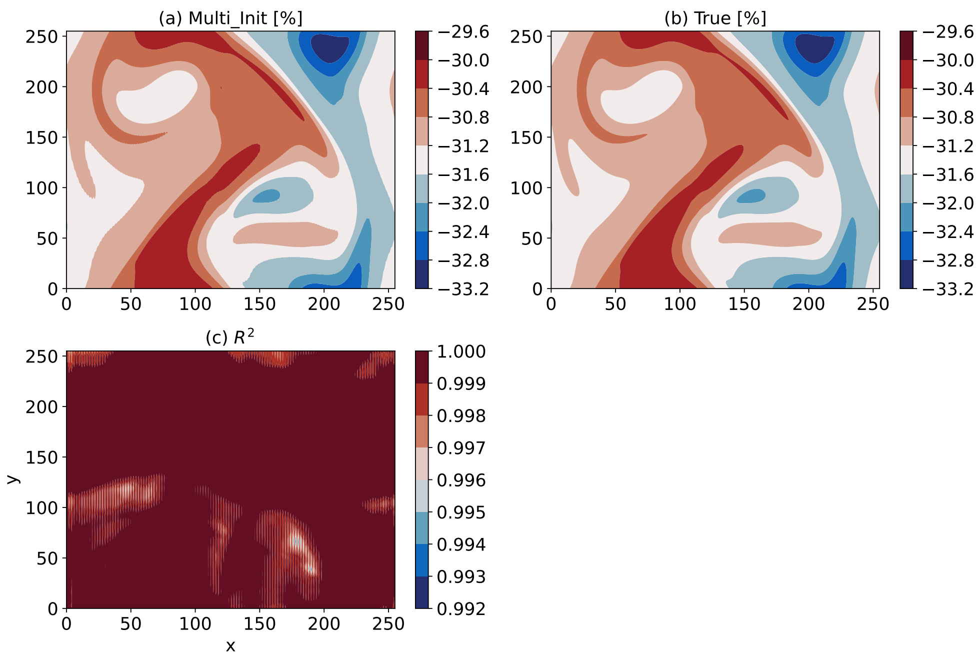

Figure 10. Multi-Init learning from the 128 x 128 resolutions from PR-DNS simulations with Init1, Init2, and Init4, test- ing on 256 x 256 resolution in terms of the time snapshot of predictions (a) and the ground truth (b), along with the R^2 accuracy (c).

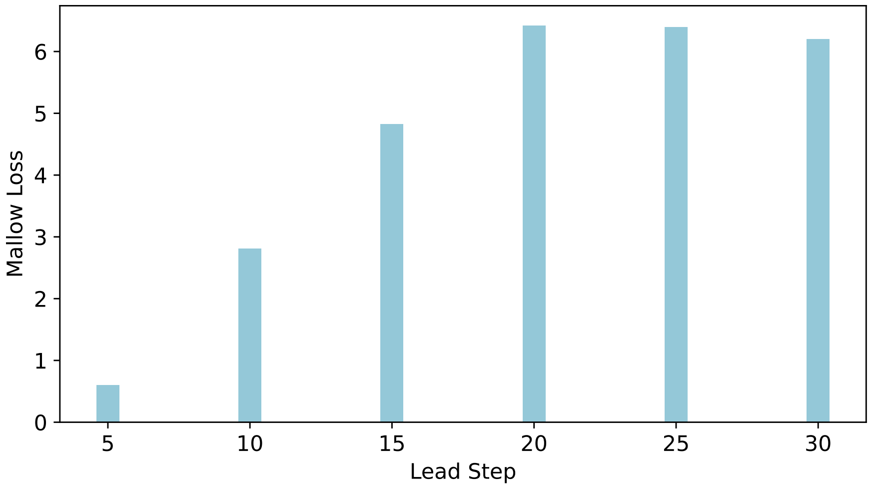

Figure 11. The Mallows loss as a function of lead step.

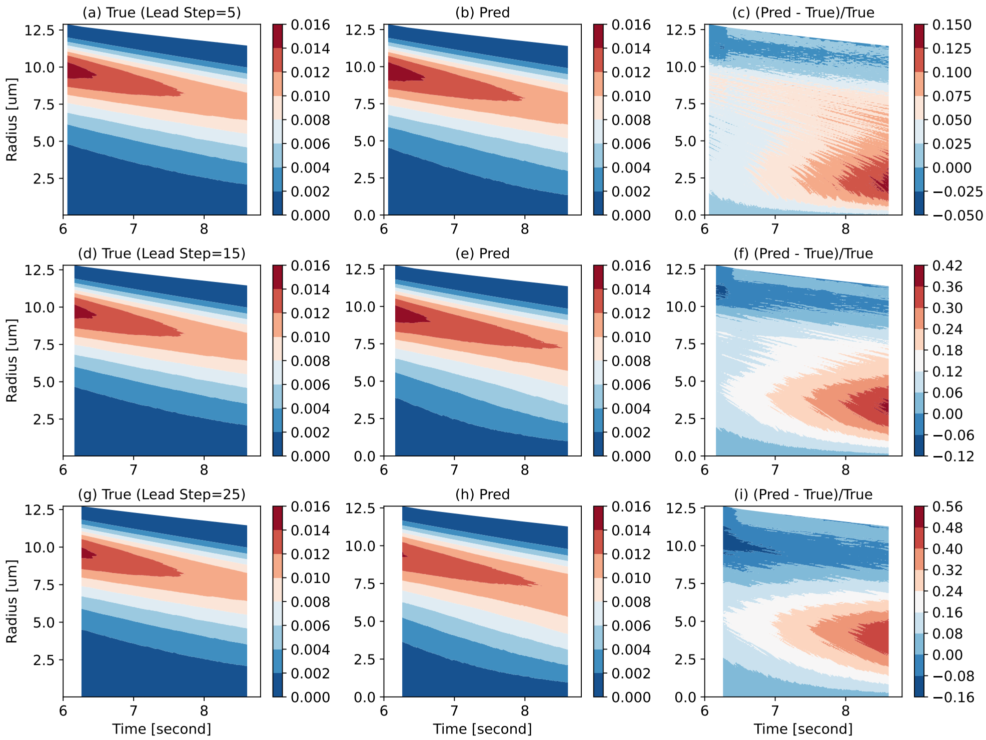

Figure 12. Evolution of droplet size distribution. The color indicates the droplet size distribution. The panels from top to bottom correspond to lead steps 5, 15, and 25, respectively. The columns from left to right depict the ground truth, predictions, and the relative errors between predictions and ground truth.

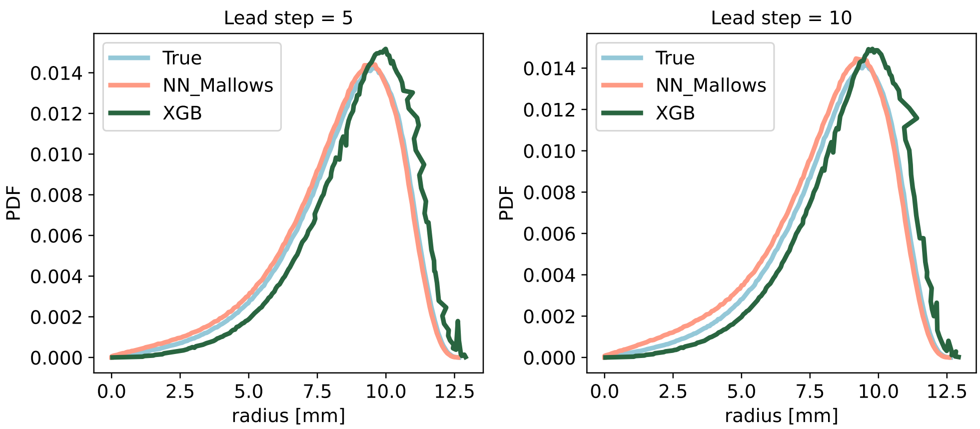

Figure 13. Comparing the regression model using the Mallows loss function with the XG-Boost method for lead steps 5 and 10.

5 Discussion and conclusion

In our previous study, we applied neural operators to construct emulators for velocity and temperature fields. Here, we extend the approach to accurately emulate the-water vapor field and the evolution of droplet size distributions using high-fidelity PR-DNS simulations. Consistent with prior results, the emulators achieve high prediction skill while significantly reducing computational cost compared to the explicit simulations.

Crucially, we enhance generalizability by training on multi-initialization datasets instead of focusing on a single initial condition. This approach enables the model to perform effectively under new conditions not present in the original training set. Furthermore, we utilize the innovative Mallows distance as the loss function for predicting the droplet radius distribution.

This work and our previous work demonstrate that the deep learning emulators are an essential tool because they allow to perform the extremely high-resolution PR-DNS simulations in an economical manner for larger domain sizes and different scenario simulations. This allows us to understand the key processes in the aerosol-cloud-turbulence system, such as how turbulence affects cloud properties. In addition, the larger domain emulators (such as 10m) can be incorporated into large-eddy simulations to support multi-scale simulations.

In the future work, two points are worth exploring. First, this work only builds emulators of 2D PR-DNS simulations, especially for dynamic and thermodynamic fields. For real simulations, we should develop 3D emulators, which are critical for capturing important 3D turbulent flow effects and cloud-turbulence interactions that cannot be modeled in 2D. Accurately representing these 3D simulation is vital for making realistic predictions of cloud microphysics, dynamics, and precipitation in the atmosphere. Second, because the emulation accuracy degrades with the lead time step increase, it is desirable to develop new emulator methods to constrain the loss error. Due to the explicit PDE equations in PR-DNS, we can use Physics-informed Neural Network methods to reduce the loss error by incorporating the governing equations as constraints (Raissi et al., 2019).

6 Data Availability Statement

The deep learning models can be archived at https://doi.org/10.5281/zenodo.13886572 (Zhang, 2024a). The PR-DNS time-step simulations at various initial conditions and various resolutions for training the deep learning models can be accessed at https://doi.org/10.5281/zenodo.13888516 (Zhang, 2024b). The PR-DNS models can be accessed at https://doi.org/10.5281/zenodo.10145685 (Zhang et al., 2023).

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References

Arroyo, J., & Maté, C. (2009). Forecasting histogram time series with k-nearest neighbours methods. *International Journal of Forecasting*, *25*(1), 192–207.

Atif, M., Dubey, P., Aghor, P. P., Lopez-Marrero, V., Zhang, T., Sharfuddin, A., ... Li, L. (2024). Fourier neural operators for spatiotemporal dynamics in two-dimensional turbulence. *arXiv preprint arXiv:2409.14660*.

Atif, M., López-Marrero, V., Zhang, T., Sharfuddin, A., Yu, K., Yang, J., ... Li, L. (2024). Towards accelerating particle-resolved direct numerical simulation with neural operators. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *17*(3), e11690.

Bhatnagar, S., Afshar, Y., Pan, S., Duraisamy, K., & Kaushik, S. (2019). Prediction of aerodynamic flow fields using convolutional neural networks. *Computational Mechanics*, *64*, 525–545.

Chen, S., Yau, M., & Bartello, P. (2018). Turbulence effects of collision efficiency and broadening of droplet size distribution in cumulus clouds. *Journal of the Atmospheric Sciences*, *75*(1), 203–217.

Dias, S., & Brito, P. (2015). Linear regression model with histogram-valued variables. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *8*(2), 75–113.

Fan, J., Wang, Y., Rosenfeld, D., & Liu, X. (2016). Review of aerosol–cloud interactions: Mechanisms, significance, and challenges. *Journal of the Atmospheric Sciences*, *73*(11), 4221–4252.

Gao, Z., Liu, Y., Li, X., & Lu, C. (2018). Investigation of turbulent entrainment-mixing processes with a new particle-resolved direct numerical simulation model. *Journal of Geophysical Research: Atmospheres*, *123*(4), 2194–2214.

Hourdin, F., Mauritsen, T., Gettelman, A., Golaz, J.-C., Balaji, V., Duan, Q., ... others (2017). The art and science of climate model tuning. *Bulletin of the American Meteorological Society*, *98*(3), 589–602.

Khairoutdinov, M., & Kogan, Y. (2000). A new cloud physics parameterization in a large-eddy simulation model of marine stratocumulus. *Monthly weather review*, *128*(1), 229–243.

Kochkov, D., Smith, J. A., Alieva, A., Wang, Q., Brenner, M. P., & Hoyer, S. (2021). Machine learning–accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, *118*(21), e2101784118.

Kullback, S. (1997). *Information theory and statistics*. Courier Corporation.

Kumar, B., Janetzko, F., Schumacher, J., & Shaw, R. A. (2012). Extreme responses of a coupled scalar–particle system during turbulent mixing. *New Journal of*

Physics, *14*(11), 115020.
Kumar, B., Schumacher, J., & Shaw, R. A. (2013). Cloud microphysical effects of turbulent mixing and entrainment. *Theoretical and Computational Fluid Dynamics*, *27*, 361–376.
LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2324.
Levina, E., & Bickel, P. (2001). The earth mover's distance is the Mallows distance: Some insights from statistics. In *Proceedings eighth ieee international conference on computer vision. iccv 2001* (Vol. 2, pp. 251–256).
Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2020a). Fourier neural operator for parametric partial differential equations. *arXiv preprint arXiv:2010.08895*.
Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2020b). Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*.
Liu, Y. (2019). Introduction to the special section on fast physics in climate models: Parameterization, evaluation, and observation. *Journal of Geophysical Research: Atmospheres*, *124*(15), 8631–8644.
Liu, Y., Yau, M.-K., Shima, S.-i., Lu, C., & Chen, S. (2023). Parameterization and explicit modeling of cloud microphysics: Approaches, challenges, and future directions. *Advances in Atmospheric Sciences*, *40*(5), 747–790.
Mallows, C. L. (1972). A note on asymptotic joint normality. *The Annals of Mathematical Statistics*, 508–515.
Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, 686–707.
Rogallo, R. S. (1981). *Numerical experiments in homogeneous turbulence* (Vol. 81315). National Aeronautics and Space Administration.
Simpkins, G. (2018). Aerosol–cloud interactions. *Nature Climate Change*, *8*(6), 457–457.
Yang, B., Qian, Y., Lin, G., Leung, R., & Zhang, Y. (2012). Some issues in uncertainty quantification and parameter tuning: A case study of convective parameterization scheme in the WRF regional climate model. *Atmospheric Chemistry and Physics*, *12*(5), 2409–2427.
Zhang, T. (2024a, October). *Emulator of PR-DNS: Part II, Accelerating thermodynamics and cloud droplet fields with machine learning in Particle-Resolved Direct Numerical Simulation.* Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.13886572` doi: 10.5281/zenodo.13886572
Zhang, T. (2024b, October). *Emulator of PR-DNS: Part II, dataset for training the emulator of thermodynamics and cloud droplet fields in Particle-Resolved Direct Numerical Simulation.* Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.13888516` doi: 10.5281/zenodo.13888516
Zhang, T., Li, L., Lopez-Marrero, V., Lin, M., Liu, Y., Fan, Y., ... Atif, M. (2023, November). *tzhang-ccs/pr_dns_base: pr_dns_cuda_v1.* [Software], Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.10145685` doi: 10.5281/zenodo.10145685
Zhang, T., Li, L., López-Marrero, V., Lin, M., Liu, Y., Yang, F., ... Atif, M. (2024). Emulator of PR-DNS: Accelerating dynamical fields with neural operators in particle-resolved direct numerical simulation. *Journal of Advances in Modeling Earth Systems*, *16*(2), e2023MS003898.

Emulator of PR-DNS: Part II, Accelerating thermodynamics and cloud droplet fields with machine learning in Particle-Resolved Direct Numerical Simulation

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Key Points:

- The Fourier Neural Operator can effectively emulate the moist process, expanding up our prior work for emulating dynamics and temperature.
- The multi-initial learning method enhances the generalizability across diverse initial conditions and super resolutions.
- The droplet radius distribution can be learned using the Mallows distance.

Plain Language Summary

Understanding the aerosol-cloud-turbulence interaction is crucial for climate and weather prediction. However, simulating these processes using extra-high-resolution particle-resolved direct numerical simulations is computationally expensive. In our previous work, we demonstrated that using machine learning emulators can significantly reduce computational costs while maintaining high accuracy. In this study, we extend the emulator to thermodynamic fields in two dimensions and droplet behavior in three dimensions. For the emulator of thermodynamic fields, we improve the generalizability across various initial conditions and resolutions without requiring retraining the machine learning models. For the emulator of droplet fields, we introduce a novel loss function to evaluate the droplet size distribution. Our findings indicate that these machine learning models could serve as effective tools for more efficient simulation of intricate aerosol-cloud-turbulence interactions.

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Abstract

Particle-resolved direct numerical simulations (PR-DNS) are crucial for unraveling the intricate interplay of aerosol-cloud-turbulence processes. However, such models are challenged by the huge computational cost due to the extremely high resolution. Our prior work showcased that leveraging machine learning emulators could slash computational expenses by two orders of magnitude while maintaining remarkable precision for dynamic fields, and exhibited generalizability across diverse initial conditions and at super-resolution scales without retraining the emulators. Building upon this foundation, this work extends the emulator's application to thermodynamic in the two spatial dimensions and droplet fields in the three spatial dimensions. Furthermore, to enhance the robust generalizability of the emulator for different initial values and super resolution, we introduce a novel multi-initial learning approach for the neural operator method. For the droplet fields, we introduce a novel loss function tailored to assess distribution differences using the Mallows distance, focusing particularly on droplet size distributions. Our findings indicate that the machine learning emulators hold promising potential to effectively mimic numerical PR-DNS simulations, thereby significantly advancing our understanding of the complex interactions within aerosol-cloud-turbulence processes.

1 Introduction

Understanding the intricate interplay within the aerosol-cloud-turbulence system is critical for accurately simulating the impacts of climate change and extreme weather events (Liu, 2019; Fan et al., 2016; Simpkins, 2018). Despite the advances of General Circulation Models (GCMs), Cloud-Resolving Models (CRMs), and Large Eddy Simulations (LES), uncertainties persist even at the high resolutions offered by LES. This is primarily attributed to the existing knowledge gaps and uncertainties in sub-grid parameterizations. Even for sub-LES scales. critical processes such as turbulence-microphysics interactions, turbulent entrainment-mixing, and droplet clustering are either inadequately represented or not at all in these models (Liu et al., 2023). This limitation significantly hampers the advancement in simulating the aerosol-cloud-turbulence system (Hourdin et al., 2017; Khairoutdinov & Kogan, 2000; Yang et al., 2012).

Particle-resolved Direct Numerical Simulation (PR-DNS) eliminates the need for sub-grid parameterizations (Chen et al., 2018; Liu, 2019). It explicitly resolves the smallest turbulent eddies for fluid dynamics and thermodynamics, as well as individually tracks the motion and growth of each droplet (Kumar et al., 2012, 2013). Within our current PR-DNS model, we employ the point-particle approximation, including essential droplet properties such as radius, position, and velocity. The droplet behavior is intricately intertwined with the dynamics and thermodynamics of the surrounding turbulent fluid (Gao et al., 2018). The PR-DNS serves as a unique and valuable tool for advancing our fundamental understanding of the process-level process of the aerosol-cloud-turbulence system. However, the extreme high resolution and tracking of individual droplets impose a significant computational cost. This limitation presently restricts its utility to simulations within smaller domains and a limited number of physical processes (Gao et al., 2018).

Recently, a line of studies use deep neural networks to build emulators to simulate turbulent fluid flows, which can significantly reduce the computational cost compared with traditional numerical solvers (Bhatnagar et al., 2019; Kochkov et al., 2021). However, neural network-based solvers often lack generalizability due to their finite-dimensional mapping. The neural operators, especially the Fourier Neural Operator, do the infinite-dimensional mapping, which map inputs to the Fourier space, learn the transformations, and subsequently reverse them back to the spatial domain (Li et al., 2020a, 2020b). Our previous research demonstrates that the neural operator emulators can achieve high accuracy, and significantly reduce computational cost by two orders of magnitude, and exhibit generalization across diverse initial conditions and super-resolution simulation without retraining the machine learning models. However, our findings have primarily focused on fluid dynamics and the temperature field (Zhang et al., 2024; Atif, López-Marrero, et al., 2024; Atif, Dubey, et al., 2024).

In this study, we expand upon our prior research to encompass the moist process, with foci on both water vapor/ supersaturation fields and microphysics/droplet size distributions. In terms of generalizability across different initial conditions and super-resolution simulations, our previous work did not provide robust generalizability because the fundamental model was trained on only a single set of initial conditions from the PR-DNS simulation. This narrow training approach could not ensure optimal performance on unseen data arising from different initial conditions. To achieve truly robust generalizability, models need to be trained on diverse PR-DNS simulations spanning a wider range of initial conditions. We thus introduce a novel multi-initial learning approach wherein our neural operator emulators are trained using multiple PR-DNS simulations with diverse initial conditions.

Furthermore, to learn the droplet size distribution, FNO cannot be directly applied. The primary challenge lies in devising a suitable loss function to measure the distance between the predicted and true distributions. While Kullback-Leibler (KL) divergence is commonly used to measure differences between probability distributions, it requires the two distributions to have identical bin structure (Kullback, 1997). However, the corresponding bin placements in the true distributions from PR-DNS and those from emulators may not align perfectly. Therefore, we design a novel loss function customized to assess distribution disparities using the Mallows distance (Levina & Bickel, 2001). Our results indicate that the new neural operator demonstrates high accuracy in both Euler dynamics, thermodynamics, and Lagrangian droplet radius distributions. This tool is pivotal for advancing PR-DNS in larger domain simulations with reduced computational costs, thereby playing a crucial role in deepening our understanding of the intricate interactions within aerosol-cloud-turbulence systems.

The remainder of the paper is structured as follows. Section 2 presents the PR-DNS model and the simulation data utilized for training the emulators. Section 3 outlines the methodologies employed, including the neural operator, multi-initial learning, and the use of the Mallows distance as a loss function for learning the droplet radius distribution. Section 4 delves into the performance evaluation of the methods. Conclusions and discussions are provided in Section 5.

2 Model and Data

The PR-DNS model incorporates the Eulerian field representation of turbulence near the Kolmogorov microscale and Lagrangian droplet tracking. Comprehensive details regarding the PR-DNS can be found in (Gao et al., 2018) and (Zhang et al., 2024). Here, we provide a brief overview of the key equations.

The Eulerian dynamic field is the incompressible Boussinesq fluid system, described by

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho_0}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}_b + \mathbf{f}_e \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{1b}$$

where $\mathbf{u}$ is the velocity field, p is the pressure field, ν is the kinematic viscosity, ρ_0 is the density of dry air, $\mathbf{f}_b$ and $\mathbf{f}_e$ are the buoyancy and external forcing, respectively.

The Eulerian thermodynamic fields, namely the temperature and vapor mixing ratio, are given by

$$\partial_t T + (\mathbf{u} \cdot \nabla)T = \frac{L_h}{c_p} C_d + \mu_T \nabla^2 T \tag{2}$$

$$\partial_t q_v + (\mathbf{u} \cdot \nabla)q_v = -C_d + \mu_v \nabla^2 q_v \tag{3}$$

where L_h is the latent heat of water vapor condensation, c_p is the specific heat, μ_T and μ_v are the molecular diffusivity for temperature and water vapor, respectively. The condensation rate C_d depends on the droplet population, thereby establishing a coupling between the Eulerian field and the Lagrangian droplets.

For the Lagrangian process, the motion of droplets is calculated by,

$$\frac{\mathrm{d}\mathbf{X}_i(t)}{\mathrm{d}t} = \mathbf{V}_i(t) \tag{4a}$$

$$\frac{\mathrm{d}\mathbf{V}_i(t)}{\mathrm{d}t} = \frac{1}{\tau_p}\left[\mathbf{u}\left(\mathbf{X}_i, t\right) - \mathbf{V}_i(t)\right] + \mathbf{g} \tag{4b}$$

where $X_i(t)$ and $V_i(t)$ are the position and velocity of the ith droplet at time t. The droplet velocity is influenced by turbulence. τ_p is the droplet response time, which measures the inertial effect. The gravity term $\mathbf{g}$ represents the sedimentation, which affects the vertical motion of droplets.

The growth of droplet condensation is described

$$R_i(t)\frac{\mathrm{d}R_i(t)}{\mathrm{d}t} = A \cdot S\left(\mathbf{X}_i, t\right) \tag{5}$$

where $R_i(t)$ represents the radius of the ith droplet at time t, A is a function of temperature and pressure. Further details can be found in (Gao et al., 2018). The supersaturation $S(X, t)$ is calculated as

$$S(\mathbf{X}, t) = \frac{q_v(\mathbf{X}, t)}{q_{v,S}(\mathbf{X}, t)} - 1 \tag{6}$$

where $q_{v,S}$ is the corresponding saturation water vapor mixing ratio.

We leverage the PR-DNS simulations to generate high-fidelity dynamic, thermodynamic and droplet dataset. In the PR-DNS simulations to learn thermodynamics fields, the physical domain is set to be $0.512 \times 0.512m^2$ in two spatial dimensions with doublly periodic boundary conditions. The training datasets are generated with 128 x 128 grid resolution, corresponding to a grid spacing of 4 mm. Additionally, to improve robust generalizability, we conduct simulations with 3 different initial velocity conditions at 128×128 grid resolution for training, reserving an additional case for testing. To evaluate the performance of predicting higher resolutions using the lower 128×128 training model, the test simulation resolution is 256×256, corresponding to a finer grid size of 2mm. The average time step size is 0.003s, satisfying the Courant–Friedrichs–Lewy stability criterion. The initial turbulent Reynolds and Rayleigh numbers are 6,827 and 40,227, respectively. Each simulation is run for 6,000 time steps. For a single initial condition, we use the first 4000 time steps of the simulation for training. For multiple initial conditions, we combine the 12,000 time steps across three simulations with distinct initial conditions for training. In each case, the last 2000 time steps are reserved for model validation. In the PR-DNS simulations to learn the droplet size distribution, we use the three-dimensional spatial domain. Further details can be found in (Gao et al., 2018). The output time interval is 0.01s. A total of 863 output steps are generated, with the first 600 steps is used to train the emulator of the droplet size distribution, and the rest for testing.

3 Method

3.1 Fourier Neural Operator

For dynamic and thermodynamic variables, like velocity, temperature and water vapor, the emulator could be developed using conventional deep learning methods, such as Convolutional Neural Networks (CNN, LeCun et al. (1998)), which utilize the current timestep as input to predict their values in the subsequent timestep. However, this approach would have limitations. Using the CNN-like methods, the relationship between the input and output is predetermined by the network architecture, as illustrated in Figure 1 (a). This method would not be adaptable for scenarios where the output resolution varies, like in the case of super resolution.

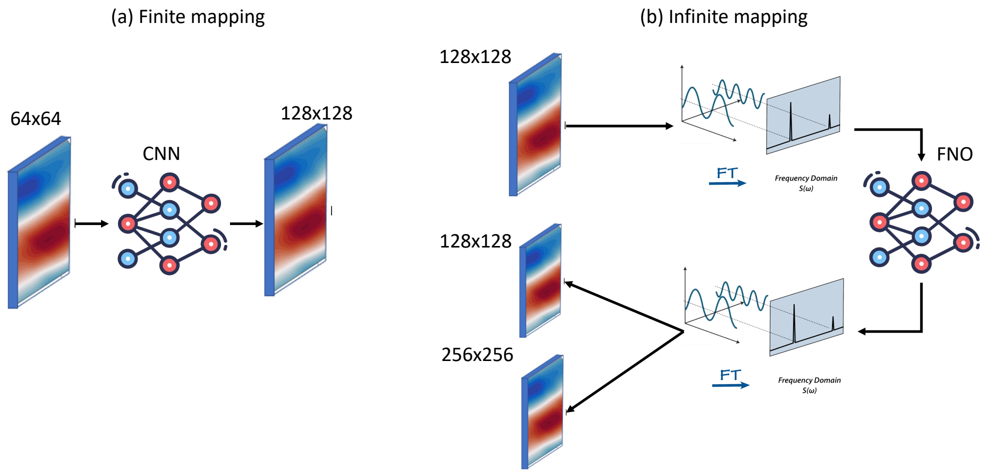

Figure 1. Differences in deep learning between finite mapping (a) and infinite mapping (b).

Instead, the Fourier Neural Operator (FNO) offers enhanced generalizability and the flexibility to adjust the output resolution. It transforms the spatial field into the Fourier domain. Subsequently, the neural network directly learns the mappings in Fourier space rather than in spatial space. Finally, it translates these representations back into the spatial domain, as shown in Figure 1 (b). Our previous work demonstrated that the FNO exhibited higher accuracy and efficiency compared to the finite mapping method for emulating both velocity and temperature fields (Zhang et al., 2024).

Details of the FNO algorithm can be found in Li et al. (2020a). Here, we provide a brief introduction to this algorithm. The FNO learning mapping equation is given by

$$\mathcal{G}_\theta = Q \circ \sigma_l \left(\mathcal{K}_l + W_l\right) \circ \cdots \circ \sigma_0 \left(\mathcal{K}_0 + W_0\right) \circ P \tag{7}$$

The neural operator can be decomposed into three components:

1. Encoder (P): It maps input spatial fields into a high-dimensional latent representation.
2. Fourier layers ($\mathcal{F}$): It is the foundational component of FNO, consisting of the integral operator (K_l) and the linear operator (W_l), where $l = 0, 1, 2, 3$, denoting the four Fourier layers.
3. Decoder (Q): It projects the Fourier output back to the original spatial field.

Both P and Q are shallow fully connected neural networks. In the integral operator (K_l), the input lifted by the encoder P first undergoes a fast Fourier transform (FFT) into Fourier space. Within this space, higher frequency noise is removed by truncating Fourier modes above a certain threshold. In our study, we find truncating above 30 modes achieves a balanced tradeoff between performance and computational efficiency. A linear neural network is then used to learn the remaining low-pass filtered Fourier coefficients. The result is reversed to spatial domain by an inverse FFT.

The output of each Fourier layer is derived from the combination of the integral operator (K_l) and linear operator (W_l), which is the 2-dimension convolution layer with Gaussian Error Linear Unit (GELU) activation function. The output is then fed into the subsequent layer. Finally, the cumulative is projected back onto the desired data dimension by the decoder network (Q).

3.2 Learning from multi-initial simulations

In our prior study (Zhang et al., 2024), we trained the FNO using PR-DNS simulations from a single set of velocity initial conditions and then applied it to obtain predictions given different initial conditions and also for higher resolution predictions to assess generalizability. However, training FNO with a single initial condition was found to lack robustness when applying the trained model to other initial conditions, as discussed in Section 4. To ensure a more robust model, we opt to train the FNO using data from multiple PR-DNS simulations conducted under a range of initial conditions.

Specifically, we generate simulations using four distinct velocity initial conditions with a 128 x 128 grid, as illustrated in Figure 2. They are generated by the Rogallo method (Rogallo, 1981), which a random factor is incorporated to generate the different initial conditions. The FNO model is subsequently trained on simulations from the first three initial conditions, and the fourth is used to test the performance, as depicted in Figure 3. By leveraging data from diverse initial conditions during training, the FNO method aims to learn the general turbulent flow dynamics rather than specific initialization-dependent behaviors. This improves the generalizability compared to our previous single-initialization FNO methodology, as will be shown in Section 4.1.

3.3 Predicting droplet size distribution with the Mallows loss

Cloud droplet size distribution is an important cloud microphysical property that impacts cloud processes and interactions within the climate system. PR-DNS can explicitly resolve the time-evolving droplet size distribution. However, building such a emulator presents challenges compared to the fluid dynamics fields. Unlike spatial grid-by-grid fields, the droplet size distribution is defined as a time series histogram, representing the frequencies within discrete radius bins at each time step:

$$\begin{aligned} h_{f_t} &= \left\{\left([\underline{I},\bar{I}]_{f1},p_{f1}\right),\left([\underline{I},\bar{I}]_{f2},p_{f2}\right),...,\left([\underline{I},\bar{I}]_{fl},p_{fl}\right)\right\}_t \\ h_{g_t} &= \left\{\left([\underline{I},\bar{I}]_{g1},p_{g1}\right),\left([\underline{I},\bar{I}]_{g2},p_{g2}\right),...,\left([\underline{I},\bar{I}]_{gl},p_{gl}\right)\right\}_t, \end{aligned} \tag{8}$$

where h_{f_t} and h_{g_t} are defined as the two histograms at the t-th time step, and h_{f_t} represents predictions and h_{g_t} represents the ground truth, $[\underline{I},\bar{I}]_i$ represents the lower and upper bounds of each histogram bin, p_i is the frequency corresponding to the bin $[\underline{I},\bar{I}]_i$, and fl and gl are the number of bins in the two histograms, respectively. The cumulative probability of the two distributions f and g are w_{f0},w_{f1}, ..., w_{fl} and w_{g0},w_{g1}, ..., w_{gl}, which are calculated by

$$w_i = \sum_j^i p_j, \tag{9}$$

where $i \in [f0, f1, ..., fl]$ or $[g0, g1, ..., gl]$. We use a simple regression model to predict each bin range and its corresponding frequency, described using PyTorch code as follows:

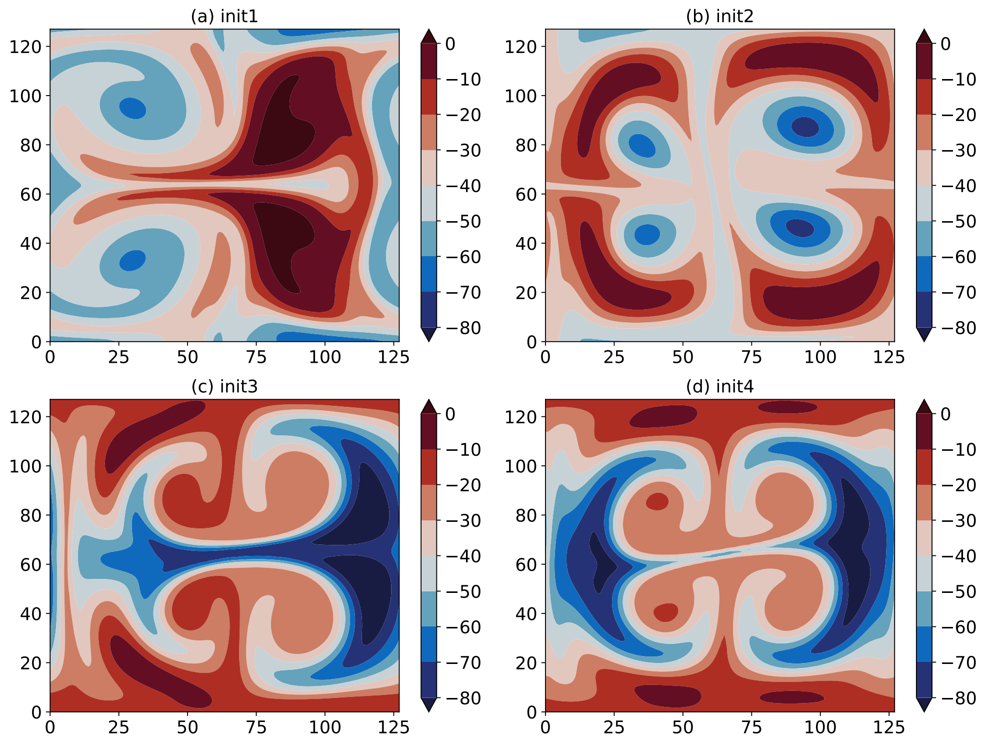

Figure 2. The time snapshot at the 100th step of supersaturation for the four different initial conditions.

```
class NN(nn.Module):
    def __init__(self,input_size, output_size):
        super(NN,self).__init__()
        self.con1 = nn.Parameter(torch.tensor(1.0))
        self.con2 = nn.Parameter(torch.tensor(-0.2))
        self.con3 = nn.Parameter(torch.ones(200))

    def forward(self, bin_input,freq_input):
        bins = bin_input[:,0,:] * self.con1 + self.con2
        freq = freq_input[:,0,:] * self.con3
        bins = torch.where(bins<0,0,bins)
        freq = torch.where(freq<=0,0,freq)
        return bins,freq
```

To build the emulator of droplet radius histogram time series, it is essential to define an appropriate loss function to measure the difference between the predicted and true distributions. While KL divergence can serve as the loss function, it may not be suitable because of the mismatched bin structures between the two distributions. Since the parameters in the algorithm above require tuning through backward optimization. During the initial iterations, the bin structures may not align well. Instead, the Mallows method can evaluate distribution dissimilarity even when the bin structures differ (Mallows, 1972). The Mallows distance $D_M(f,g)$ is defined as follows and is represented by the shaded area in Figure 4:

$$D_M(f,g) = \sqrt{\sum_{z=1}^{m} \left(H_f^{-1}(z) - H_g^{-1}(z)\right)^2}, \tag{10}$$

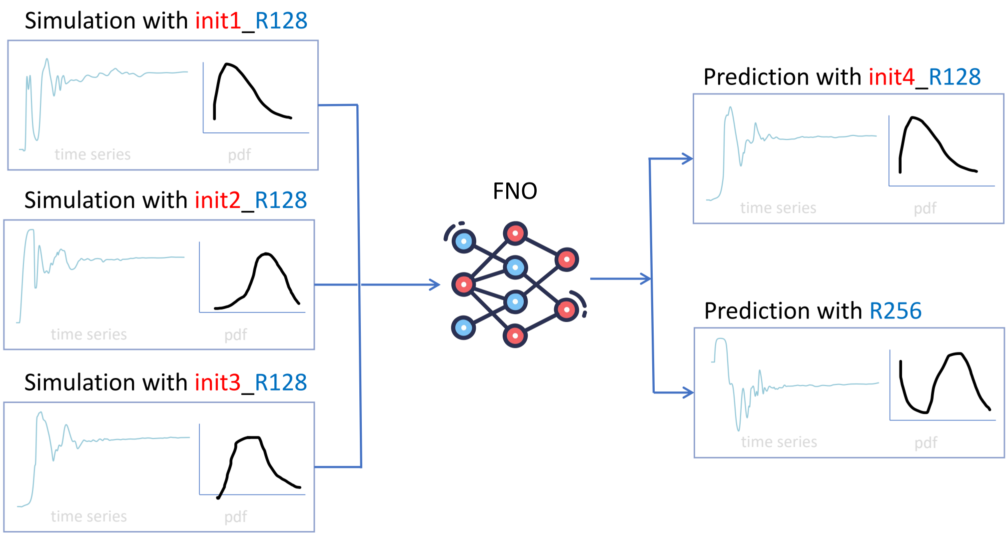

Figure 3. Illustration diagram of training FNO model with multi-initial simulations for different initial conditions and super resolution.

where $H_f^{-1}(t)$ and $H_g^{-1}(t)$ with are the inverse cumulative distribution functions of f and g, respectively, which can be defined by

$$H^{-1}(z) = \underline{I}_l + \frac{z - w_{l-1}}{w_l - w_{l-1}} \left(\bar{I}_l - \underline{I}_l\right) \\ \text{if } z \in \left[w_{l-1}, w_l\right), \tag{11}$$

It is worth noting that $H^{-1}(z)$ is a function with z as the single parameter. Both H_f^{-1} and H_g^{-1} utilize the same z value to calculate the difference. In Figure 4, z represents the values from the y-axis, obtained by combining the cumulative probabilities w for f and g, and then sorted without repetitions; m denotes the length of the z array.

Within the interval (z_{l-1}, z_l), it follows the uniform distribution. Hence, the inverse cumulative distribution can be expressed as:

$$H^{-1}(t) = c_l + r_l(2t - 1) \quad z_{l-1} \leq t \leq z_l \tag{12}$$

where the c_l and r_l are the center and radius of each bin l. The center of a bin refers to the midpoint of the bin. The radius of a bin is half of the width of the bin. To obtain the center and radius, the lower bound ($\underline{I}$) and upper bound ($\bar{I}$) for the interval (z_{l-1}, z_l) can be derived from the inverse cumulative function $H^{-1}(z_{l-1})$ and $H^{-1}(z_l)$, respectively. They can subsequently be calculated as follows

$$c_l = \frac{\underline{I}_l + \bar{I}_l}{2} \quad \text{and} \quad r_l = \frac{\bar{I}_l - \underline{I}_l}{2} \tag{13}$$

The Mallows distance in Eq. 10 can be rewritten as follows according to (Arroyo & Maté, 2009) and (Dias & Brito, 2015)

$$
\begin{aligned}
D_M(f,g) &= \sum_{l=1}^{m} \sqrt{\int_{z_{l-1}}^{z_l} \left(H_f^{-1}(t) - H_g^{-1}(t)\right)^2 \,\mathrm{d}t} \\
&= \sum_{l=1}^{m} p_l \sqrt{\int_0^1 [(c_{lf} + r_{lf}(2t-1)) - (c_{lg} + r_{lg}(2t-1))]^2 \,\mathrm{d}t} \\
&= \sum_{l=1}^{m} p_l \sqrt{[(c_{lf} - c_{lg})^2 + \frac{1}{3}(r_{lf} - r_{lg})^2]}
\end{aligned} \tag{14}
$$

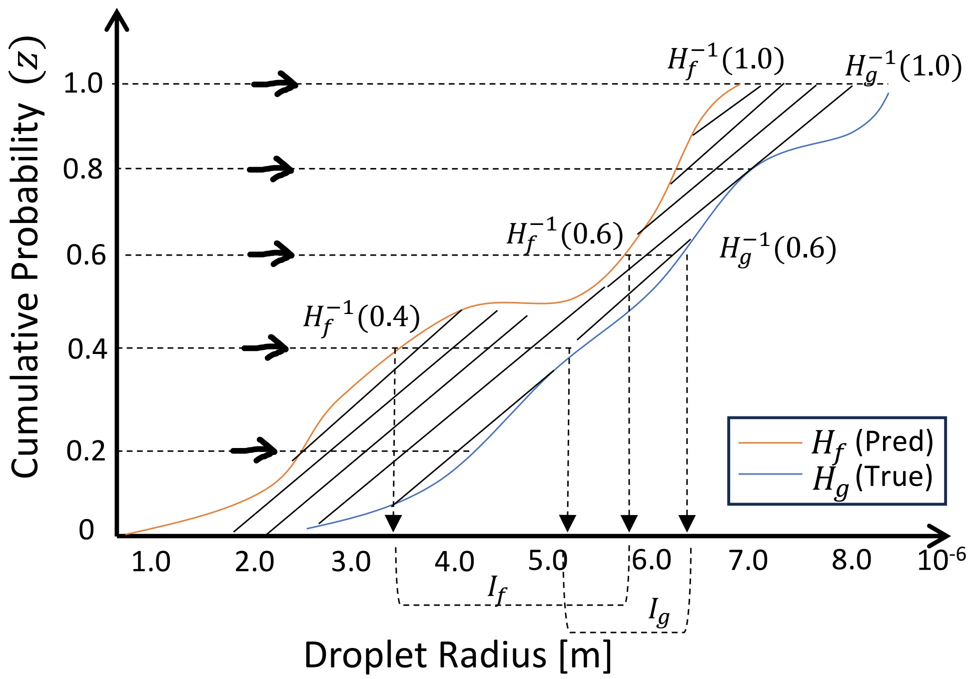

Figure 4. The Mallows distance quantifies the difference in droplet of the predicted and true cumulative radius distribution. The shaded area $\sqrt{\sum_{z=1}^{m}\left(H_f^{-1}(z) - H_g^{-1}(z)\right)^2}$ represents this distance.

4 Results

In this section, we assess the emulator performance of Eulerian thermodynamic fields and Lagrangian droplet radius distribution in predicting cloud microphysical properties.

4.1 Thermodynamics

4.1.1 Accuracy

In our prior research (Zhang et al., 2024), we conducted a comparative evaluation of three machine learning approaches for predicting dynamic and temperature fields: the UNet, ResNet, and FNO. The FNO exhibited the lowest prediction errors while achieving powerful generalizability. Therefore, in this current work, we focus on results using the FNO model alone.

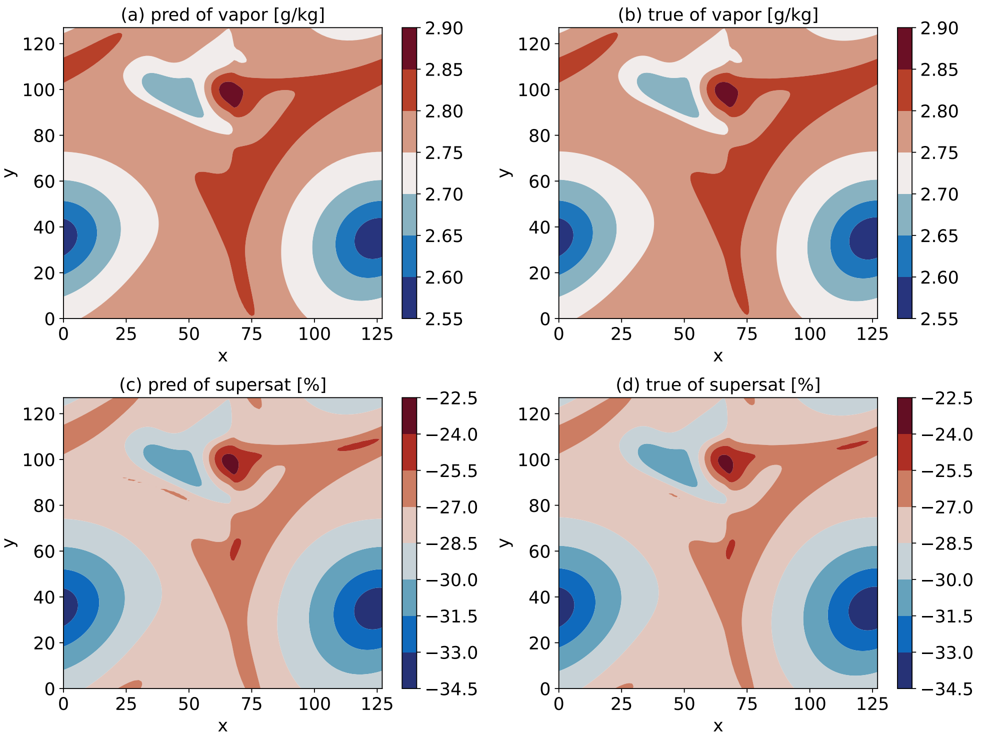

Figure 5. Snapshot predictions (a,c) and ground truth (b,d) of mixing ratio and supersaturation at the 5900th time step (equivalent to the 1900th time step in the testing dataset. The following figures about snapshot are all from the 5900th time step) with a resolution of 128 × 128.

Figure 5 displays the two-dimensional (2D) spatial maps of mixing ratio and supersaturation at a 128 x 128 resolution from a single time snapshot, as examples. The figure shows that the FNO predictions very closely resemble the ground truth obtained from the PR-DNS simulation for the two thermodynamic variables.

To further quantitatively evaluate the accuracy of the deep learning approach, Figure 6 show the coefficient of determination R^2 between FNO predictions and ground truth, evaluated over the entire testing dataset on a grid point by grid point basis. It represents the fit goodness through the proportion of variance. The value 1.0 of R^2 represents a perfect fit, and the value can range from negative to 1.0. The higher the value, the better the performance. The R^2 accuracy metric is calculated as follows,

$$R^2(x_i, y_j) = 1 - \frac{\sum_{k=1}^{T} \left(u(x_i, y_j, t_k) - \hat{u}(x_i, y_j, t_k)\right)^2}{\sum_{k=1}^{T} \left(u(x_i, y_j, t_k) - \bar{u}(x_i, y_j)\right)^2} \tag{15}$$

where x_i and y_j represent grid points for the spatial coordinates in the 2D space. T is the total length of the testing dataset (corresponding to the total number of testing time steps $\{t_k\}_{k=1}^{T}$ of time coordinates). $\hat{u}$ is the prediction by deep learning models, u denotes the corresponding truth, and $\bar{u}$ is the temporal average of u.

The spatial maps and error analyses demonstrate that the FNO emulators can accurately reproduce the original PR-DNS simulation in terms of the mixing ratio and supersaturation as well. Because the patterns of mixing ratio and supersaturation are similar, we will focus on the supersaturation field in the subsequent analysis.

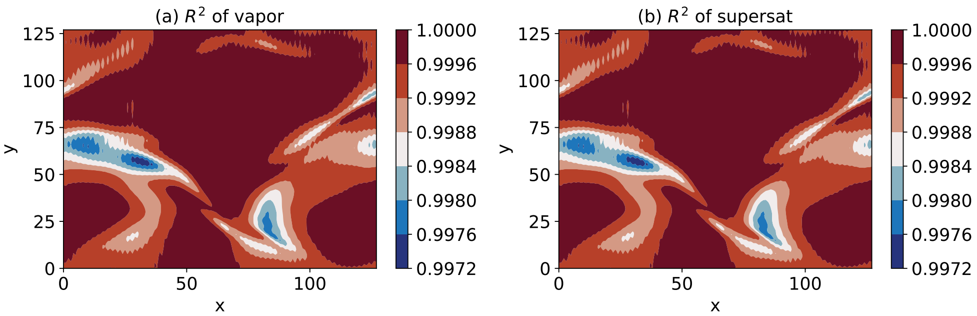

Figure 6. R^2 accuracy metric of mixing ratio (a) and supersaturation (b) between FNO predictions and the ground truth for the entire testing data set on the grid point by grid point with the resolution of 128 × 128, according to Equation 15.

4.1.2 Generalizability for different initial conditions and resolutions

While we have previously demonstrated the generalizability of our approach to different velocity initial conditions and higher resolutions, directly applying a model trained using one set of initial conditions to another set may introduce biases. We perform four PR-DNS simulations with different initial conditions, shown in Figure 2. Figure 7 (d) illustrates the case where the model trained on Init1 is applied to the Init4 conditions. The predictions exhibit a positive bias compared to the true values, suggesting the model has overfitted. Due to the magnitude similarity between init1 and init2, this bias is reduced when the model trained on init1 is applied to the init2 case, as shown in Figure 7 (b).

To address the issue, we combined data from the PR-DNS simulations using Init1, Init2 and Init3 to form a single, augmented training dataset. The FNO model is then trained using this combined dataset to learn patterns across the different data characteristics, rather than specializing to the single data. We evaluate the generalization of this novel training method on the independent Init4 condition. Figure 8 shows the greatly improved agreement between the prediction and the ground truth for the supersaturation field at the one time snapshot. This suggests that the model can effectively capture the features despite variations in the initial profiles. Figure 8 (c) confirms this method yields skillful predictions across the entire Init4 simulation based on the R^2 accuracy using the metrics defined by equation 15.

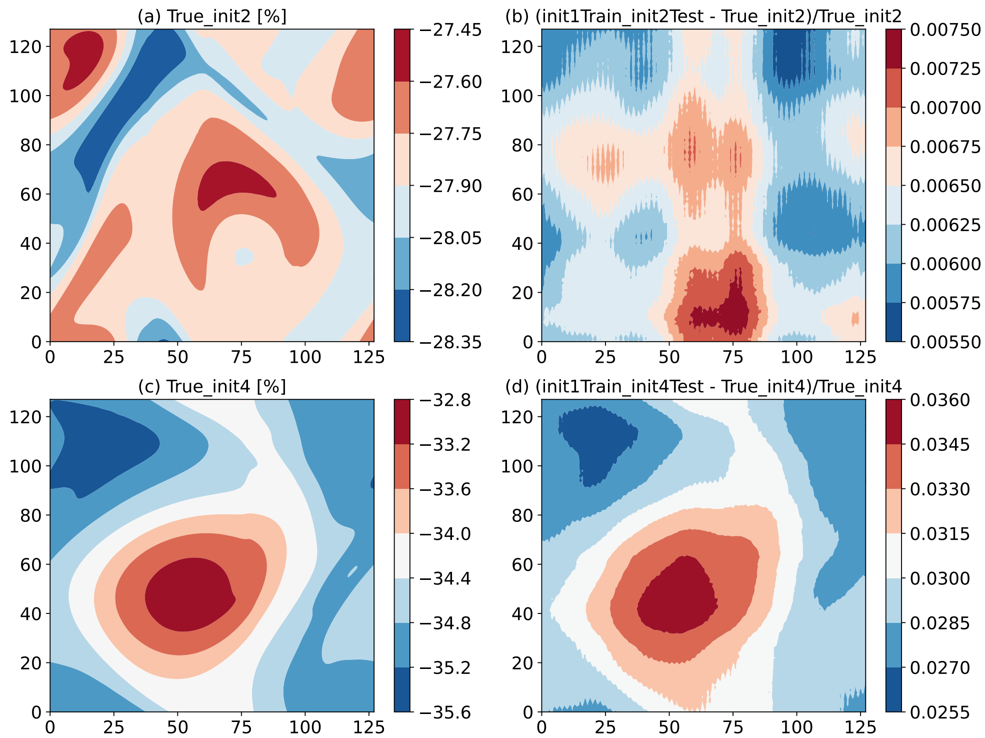

Figure 7. Testing the model trained on Init1 with Init2 and Init4, in terms of the ground truths and their relative predictive errors.

To assess the predictive capacity of super resolution using the emulator trained by the lower resolution dataset, we train the FNO model using PR-DNS simulations at a 128 x 128 resolution and apply it to predict fields at a higher resolution of 256 x 256. However, directly using the FNO model trained on one set of 128 x 128 resolution data could lead to biased predictions. In Figure 9, it is confirmed that training the FNO model using init1 at a 128 x 128 resolution and applying it to the 256 x 256 resolution results in a positive bias, as shown in Figure 9 (c). This discrepancy arises because the training data from init1 lower resolution simulation systematically exceeds the testing data from the higher resolution simulation, as illustrated in Figure 9 (a-b).

To address this challenge, we once again utilize the Multi-Init training approach. Figures 10 (a-b) shows the outcomes when employing this Multi-Init trained model to predict the 256 x 256 resolution in a single time snapshot. The predicted supersaturation field aligns closely with both the spatial patterns and the magnitude range of the actual field. Figure 10 (c) validates that this method produces accurate predictions throughout the entire Init4 simulation, as demonstrated by the R^2 accuracy metric.

4.2 Droplet Size Distribution

To assess the surrogate model's capability in predicting droplet size distribution, we employ the Mallows distance to define the loss function for the neural network surrogate model, a metric particularly suitable for comparing various histograms. Due to the subtle and gradual evolution of radius distributions over short time intervals, we extend the evaluation of predictions to longer timeframes ranging from 5 to 30 timesteps.

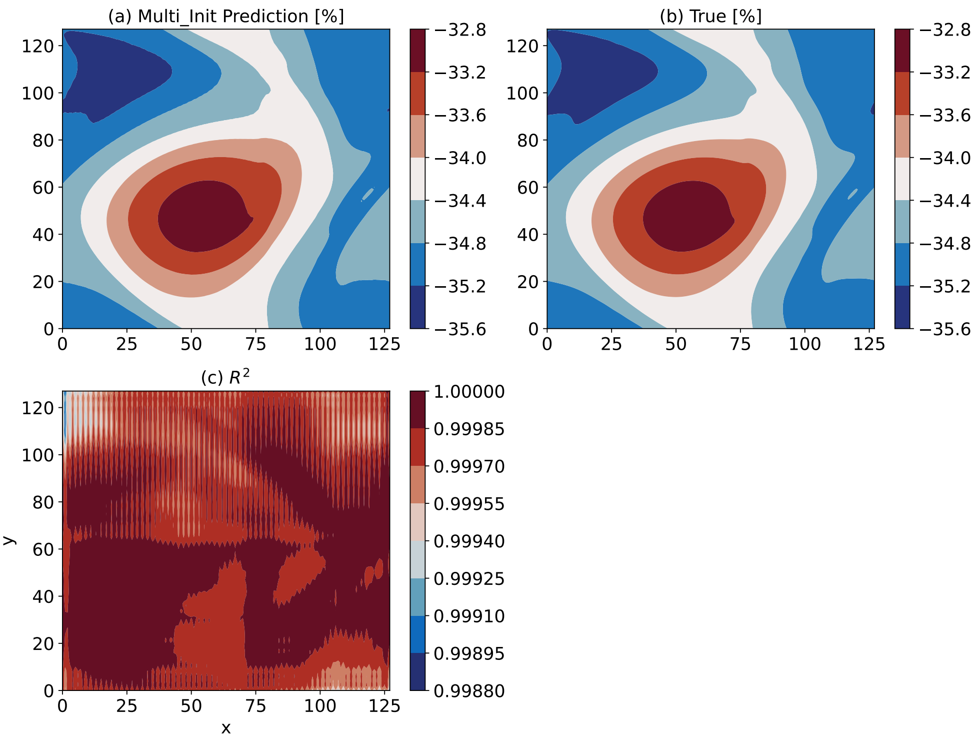

Figure 8. Multi-Init learning from the PR-DNS simulations with Init1, Init2, and Init3, testing on Init4 in terms of the time snapshot of predictions (a) and the ground truth (b), along with the R^2 accuracy (c).

Figure 11 illustrates that the Mallows loss rises as the forecast lead steps increase stabilizing after 20 steps (Here, the lead step refers to how far ahead the ML model is trying to predict. For example, if the ML model takes the current time step as input and predicts the values at the 7th subsequent step, the lead step would be 7). This indicates that the surrogate model retains its skill in simulating distribution changes, even as small uncertainties accumulate over longer time intervals. This observation is reinforced by qualitatively comparing the predicted and actual radius probability density fields at selected short and long lead times. The overall patterns remain comparable between true and predicted distributions, even out to the 30-step forecast horizon, as depicted in Figure 12.

We compare the regression method based on the Mallows loss function, as described in Section 3, with the conventional machine learning method, XGBoost. Two XGBoost models are configured: one for predicting the bins and the other for predicting the frequencies at the future step. Both XGBoost models take the bins and frequencies at the current step as input. In contrast to XGBoost, the Mallows loss function concurrently considers both aspects. Illustrated in Figure 13, the separate predictions in XGBoost result in a non-smooth distribution and significantly higher bias compared to the ground truth. Conversely, the Mallows loss function generates a smoother distribution with reduced bias.

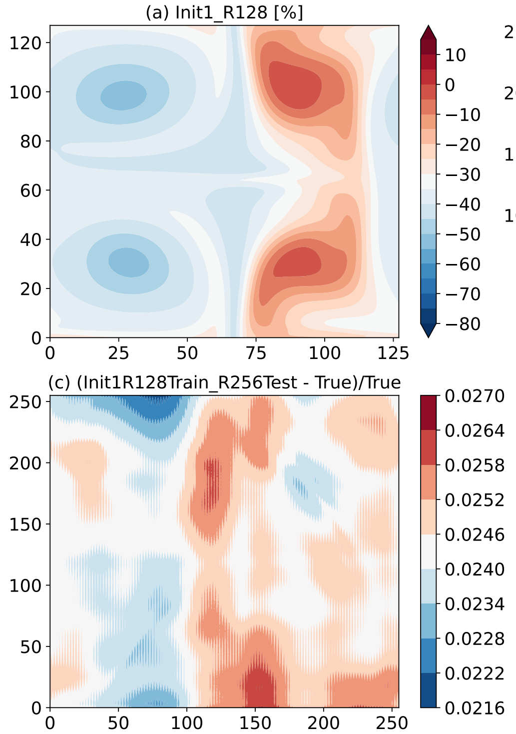

Figure 9. Testing the model trained on Init1 128 x 128 resolution with 256 x 256 resolution.

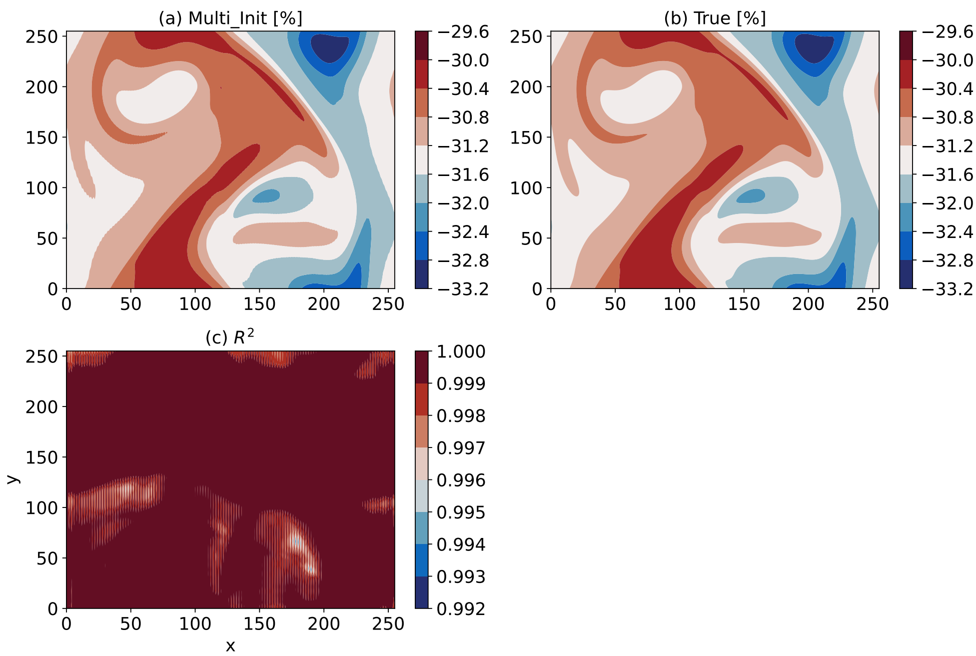

Figure 10. Multi-Init learning from the 128 x 128 resolutions from PR-DNS simulations with Init1, Init2, and Init4, test- ing on 256 x 256 resolution in terms of the time snapshot of predictions (a) and the ground truth (b), along with the R^2 accuracy (c).

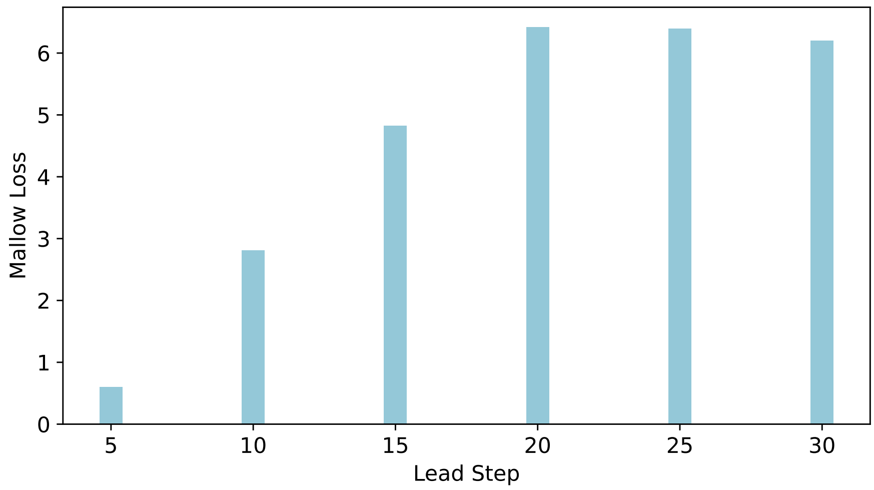

Figure 11. The Mallows loss as a function of lead step.

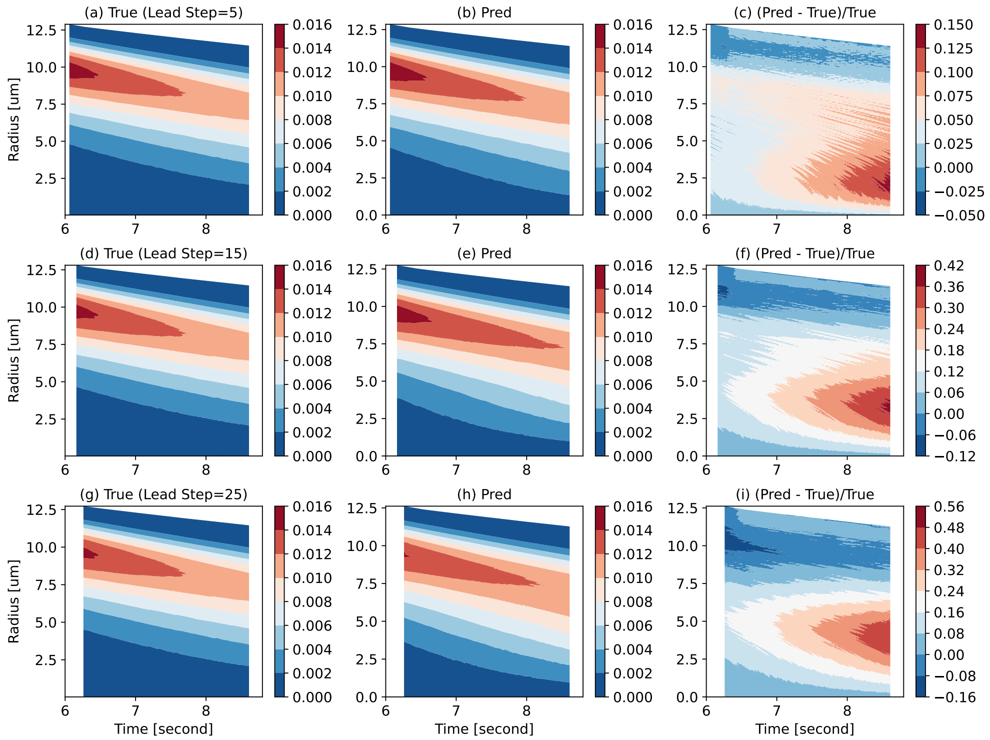

Figure 12. Evolution of droplet size distribution. The color indicates the droplet size distribution. The panels from top to bottom correspond to lead steps 5, 15, and 25, respectively. The columns from left to right depict the ground truth, predictions, and the relative errors between predictions and ground truth.

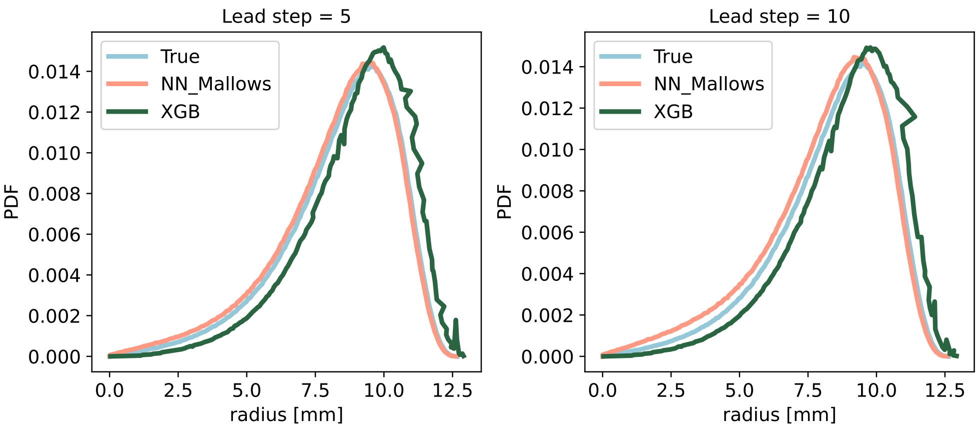

Figure 13. Comparing the regression model using the Mallows loss function with the XG-Boost method for lead steps 5 and 10.

5 Discussion and conclusion

In our previous study, we applied neural operators to construct emulators for velocity and temperature fields. Here, we extend the approach to accurately emulate thewater vapor field and the evolution of droplet size distributions using high-fidelity PR-DNS simulations. Consistent with prior results, the emulators achieve high prediction skill while significantly reducing computational cost compared to the explicit simulations.

Crucially, we enhance generalizability by training on multi-initialization datasets instead of focusing on a single initial condition. This approach enables the model to perform effectively under new conditions not present in the original training set. Furthermore, we utilize the innovative Mallows distance as the loss function for predicting the droplet radius distribution.

This work and our previous work demonstrate that the deep learning emulators are an essential tool because they allow to perform the extremely high-resolution PR-DNS simulations in an economical manner for larger domain sizes and different scenario simulations. This allows us to understand the key processes in the aerosol-cloud-turbulence system, such as how turbulence affects cloud properties. In addition, the larger domain emulators (such as 10m) can be incorporated into large-eddy simulations to support multi-scale simulations.

In the future work, two points are worth exploring. First, this work only builds emulators of 2D PR-DNS simulations, especially for dynamic and thermodynamic fields. For real simulations, we should develop 3D emulators, which are critical for capturing important 3D turbulent flow effects and cloud-turbulence interactions that cannot be modeled in 2D. Accurately representing these 3D simulation is vital for making realistic predictions of cloud microphysics, dynamics, and precipitation in the atmosphere. Second, because the emulation accuracy degrades with the lead time step increase, it is desirable to develop new emulator methods to constrain the loss error. Due to the explicit PDE equations in PR-DNS, we can use Physics-informed Neural Network methods to reduce the loss error by incorporating the governing equations as constraints (Raissi et al., 2019).

6 Data Availability Statement

The deep learning models can be archived at https://doi.org/10.5281/zenodo.13886572 (Zhang, 2024a). The PR-DNS time-step simulations at various initial conditions and various resolutions for training the deep learning models can be accessed at https://doi.org/10.5281/zenodo.13888516 (Zhang, 2024b). The PR-DNS models can be accessed at https://doi.org/10.5281/zenodo.10145685 (Zhang et al., 2023).

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References

Arroyo, J., & Maté, C. (2009). Forecasting histogram time series with k-nearest neighbours methods. *International Journal of Forecasting*, *25*(1), 192–207.

Atif, M., Dubey, P., Aghor, P. P., Lopez-Marrero, V., Zhang, T., Sharfuddin, A., ... Li, L. (2024). Fourier neural operators for spatiotemporal dynamics in two-dimensional turbulence. *arXiv preprint arXiv:2409.14660*.

Atif, M., López-Marrero, V., Zhang, T., Sharfuddin, A., Yu, K., Yang, J., ... Li, L. (2024). Towards accelerating particle-resolved direct numerical simulation with neural operators. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *17*(3), e11690.

Bhatnagar, S., Afshar, Y., Pan, S., Duraisamy, K., & Kaushik, S. (2019). Prediction of aerodynamic flow fields using convolutional neural networks. *Computational Mechanics*, *64*, 525–545.

Chen, S., Yau, M., & Bartello, P. (2018). Turbulence effects of collision efficiency and broadening of droplet size distribution in cumulus clouds. *Journal of the Atmospheric Sciences*, *75*(1), 203–217.

Dias, S., & Brito, P. (2015). Linear regression model with histogram-valued variables. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, *8*(2), 75–113.

Fan, J., Wang, Y., Rosenfeld, D., & Liu, X. (2016). Review of aerosol–cloud interactions: Mechanisms, significance, and challenges. *Journal of the Atmospheric Sciences*, *73*(11), 4221–4252.

Gao, Z., Liu, Y., Li, X., & Lu, C. (2018). Investigation of turbulent entrainment-mixing processes with a new particle-resolved direct numerical simulation model. *Journal of Geophysical Research: Atmospheres*, *123*(4), 2194–2214.

Hourdin, F., Mauritsen, T., Gettelman, A., Golaz, J.-C., Balaji, V., Duan, Q., ... others (2017). The art and science of climate model tuning. *Bulletin of the American Meteorological Society*, *98*(3), 589–602.

Khairoutdinov, M., & Kogan, Y. (2000). A new cloud physics parameterization in a large-eddy simulation model of marine stratocumulus. *Monthly weather review*, *128*(1), 229–243.

Kochkov, D., Smith, J. A., Alieva, A., Wang, Q., Brenner, M. P., & Hoyer, S. (2021). Machine learning–accelerated computational fluid dynamics. *Proceedings of the National Academy of Sciences*, *118*(21), e2101784118.

Kullback, S. (1997). *Information theory and statistics.* Courier Corporation.

Kumar, B., Janetzko, F., Schumacher, J., & Shaw, R. A. (2012). Extreme responses of a coupled scalar–particle system during turbulent mixing. *New Journal of*

Physics, *14*(11), 115020.
Kumar, B., Schumacher, J., & Shaw, R. A. (2013). Cloud microphysical effects of turbulent mixing and entrainment. *Theoretical and Computational Fluid Dynamics*, *27*, 361–376.
LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2324.
Levina, E., & Bickel, P. (2001). The earth mover's distance is the Mallows distance: Some insights from statistics. In *Proceedings eighth ieee international conference on computer vision. iccv 2001* (Vol. 2, pp. 251–256).
Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2020a). Fourier neural operator for parametric partial differential equations. *arXiv preprint arXiv:2010.08895*.
Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2020b). Neural operator: Graph kernel network for partial differential equations. *arXiv preprint arXiv:2003.03485*.
Liu, Y. (2019). Introduction to the special section on fast physics in climate models: Parameterization, evaluation, and observation. *Journal of Geophysical Research: Atmospheres*, *124*(15), 8631–8644.
Liu, Y., Yau, M.-K., Shima, S.-i., Lu, C., & Chen, S. (2023). Parameterization and explicit modeling of cloud microphysics: Approaches, challenges, and future directions. *Advances in Atmospheric Sciences*, *40*(5), 747–790.
Mallows, C. L. (1972). A note on asymptotic joint normality. *The Annals of Mathematical Statistics*, 508–515.
Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, 686–707.
Rogallo, R. S. (1981). *Numerical experiments in homogeneous turbulence* (Vol. 81315). National Aeronautics and Space Administration.
Simpkins, G. (2018). Aerosol–cloud interactions. *Nature Climate Change*, *8*(6), 457–457.
Yang, B., Qian, Y., Lin, G., Leung, R., & Zhang, Y. (2012). Some issues in uncertainty quantification and parameter tuning: A case study of convective parameterization scheme in the WRF regional climate model. *Atmospheric Chemistry and Physics*, *12*(5), 2409–2427.
Zhang, T. (2024a, October). *Emulator of PR-DNS: Part II, Accelerating thermodynamics and cloud droplet fields with machine learning in Particle-Resolved Direct Numerical Simulation.* Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.13886572` doi: 10.5281/zenodo.13886572
Zhang, T. (2024b, October). *Emulator of PR-DNS: Part II, dataset for training the emulator of thermodynamics and cloud droplet fields in Particle-Resolved Direct Numerical Simulation.* Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.13888516` doi: 10.5281/zenodo.13888516
Zhang, T., Li, L., Lopez-Marrero, V., Lin, M., Liu, Y., Fan, Y., ... Atif, M. (2023, November). *tzhang-ccs/pr_dns_base: pr_dns_cuda_v1.* [Software], Zenodo. Retrieved from `https://doi.org/10.5281/zenodo.10145685` doi: 10.5281/zenodo.10145685
Zhang, T., Li, L., López-Marrero, V., Lin, M., Liu, Y., Yang, F., ... Atif, M. (2024). Emulator of PR-DNS: Accelerating dynamical fields with neural operators in particle-resolved direct numerical simulation. *Journal of Advances in Modeling Earth Systems*, *16*(2), e2023MS003898.